

Model 0

```
tf.keras.backend.clear_session ()

# initializing the neural networks
model = Sequential ()

# Adding a dense layer with soft max activation function, with the input
dimension determined by the shape of the training data.
Model.add(Dense (num_classes, activation = 'Soft max', input_dim=x_train.
shape [1]))

#Printing the summary.
Model.summary ()

batch_size = x_train.shape [0]
epochs = 10

# time.time () returns the time in seconds since Thu Jan 1 00:00:00 1970.
start = time.time ()

# Fitting the model.
history = model.fit (
    x_train, y_train,
    validation_data= (x_val, y_val),
    batch_size=batch_size,
    epochs=epochs
)

# time.time () returns the time in seconds since Thu Jan 1 00:00:00 1970.
end = time.time ()
```

```

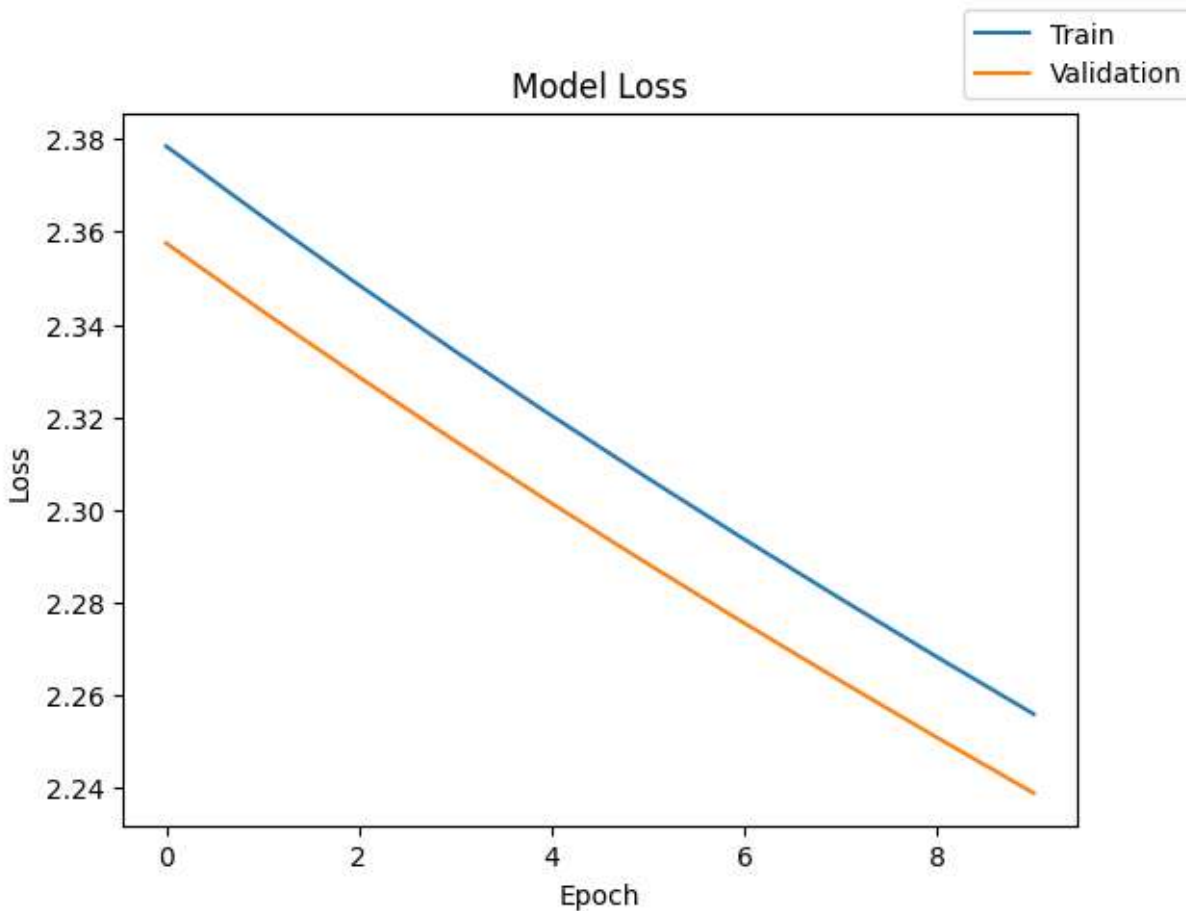
Epoch 1/10
1/1 — 1s 1s/step - accuracy: 0.0933 - loss: 2.3785 - val_accuracy: 0.1019 - val_loss: 2.3576
Epoch 2/10
1/1 — 0s 360ms/step - accuracy: 0.0996 - loss: 2.3633 - val_accuracy: 0.1081 - val_loss: 2.3429
Epoch 3/10
1/1 — 0s 345ms/step - accuracy: 0.1053 - loss: 2.3485 - val_accuracy: 0.1159 - val_loss: 2.3286
Epoch 4/10
1/1 — 0s 369ms/step - accuracy: 0.1115 - loss: 2.3342 - val_accuracy: 0.1224 - val_loss: 2.3148
Epoch 5/10
1/1 — 1s 684ms/step - accuracy: 0.1180 - loss: 2.3203 - val_accuracy: 0.1286 - val_loss: 2.3014
Epoch 6/10
1/1 — 1s 887ms/step - accuracy: 0.1244 - loss: 2.3068 - val_accuracy: 0.1342 - val_loss: 2.2882
Epoch 7/10
1/1 — 1s 1s/step - accuracy: 0.1313 - loss: 2.2936 - val_accuracy: 0.1408 - val_loss: 2.2755
Epoch 8/10
1/1 — 1s 646ms/step - accuracy: 0.1380 - loss: 2.2807 - val_accuracy: 0.1480 - val_loss: 2.2630
Epoch 9/10
1/1 — 1s 620ms/step - accuracy: 0.1455 - loss: 2.2682 - val_accuracy: 0.1554 - val_loss: 2.2508
Epoch 10/10
1/1 — 0s 463ms/step - accuracy: 0.1527 - loss: 2.2559 - val_accuracy: 0.1628 - val_loss: 2.2388

```

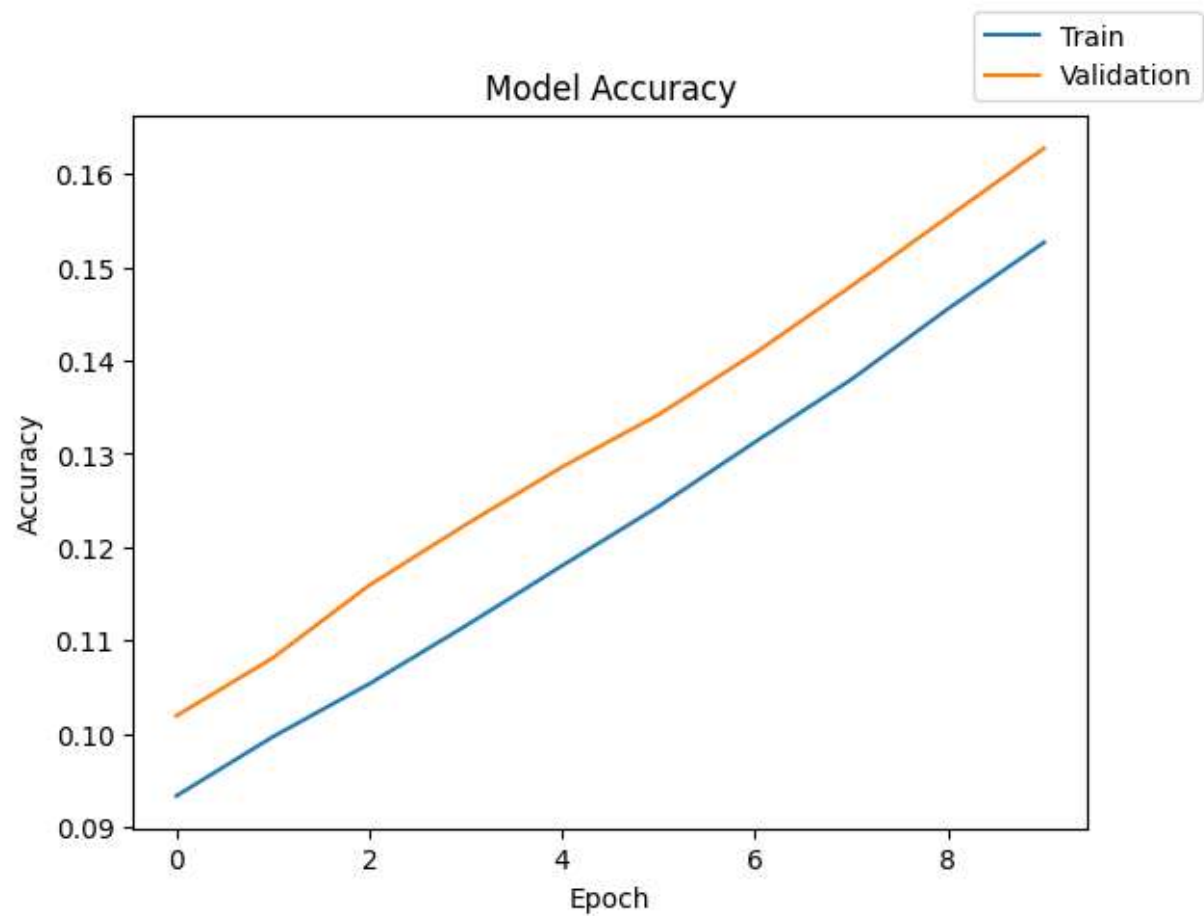
Print ("Time taken in seconds ", end-start)

Time taken in seconds 7.029578685760498

Plot (history, 'loss')



Plot (history, 'accuracy')



```
Results.loc [0] = [0,'-', '-',10,50000, history.history["loss"] [-1], history.  
History ["Val_ loss"] [-1], history.history ["accuracy"] [-1], history. History  
["Val_ accuracy"] [-1], round(end-start,2)]
```

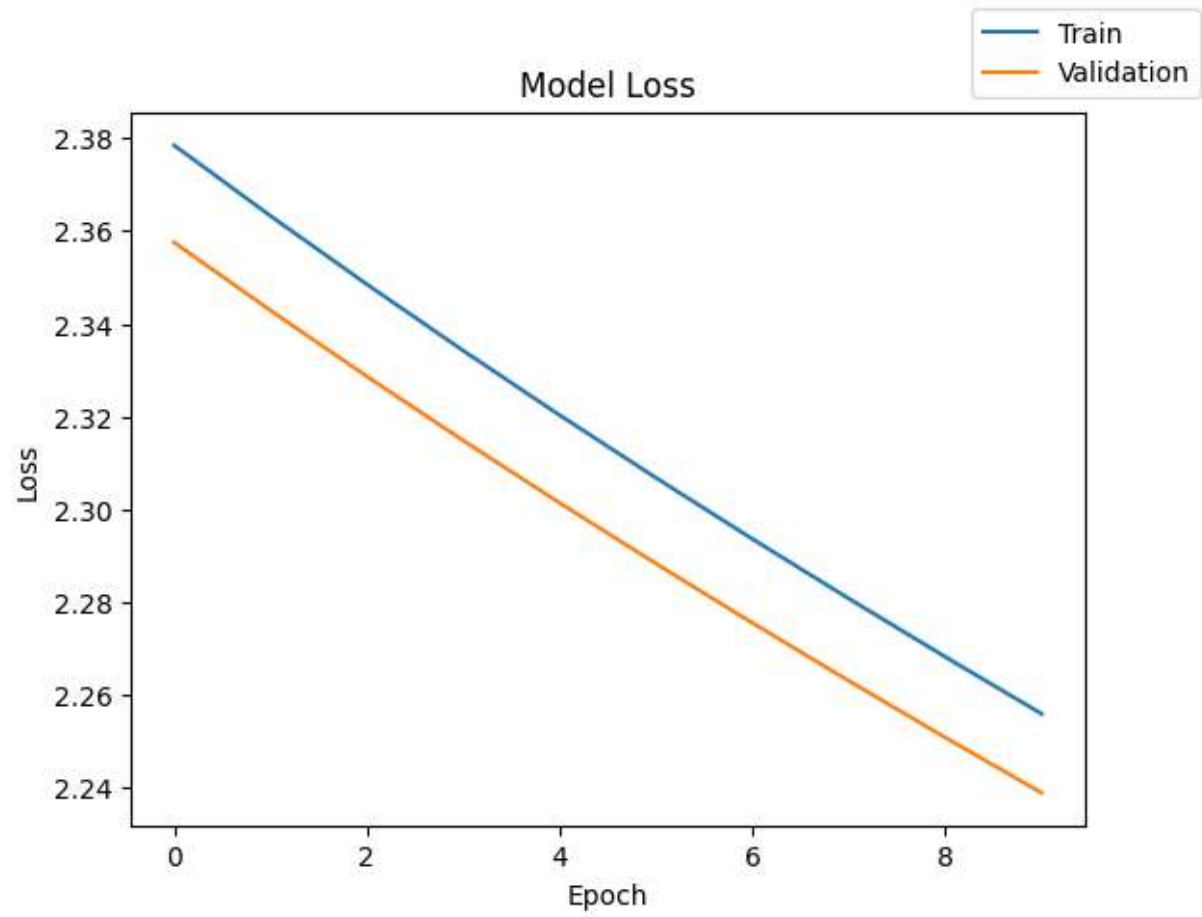
results

# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	10	50000	2.255868	2.238811	0.1527	0.1628	7.03

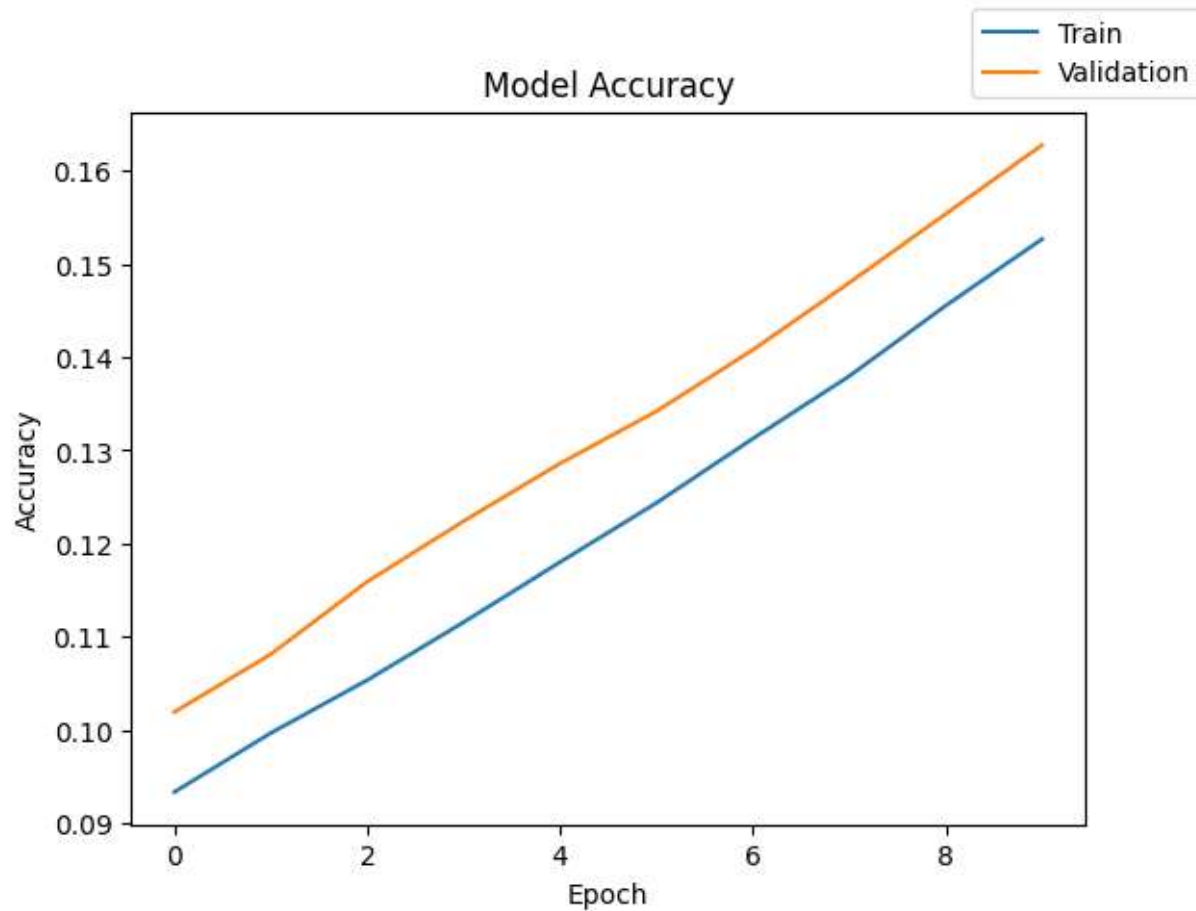
Model 1

```
Print ("Time taken in seconds ", end-start)  
Time taken in Seconds 22. 816673040390015
```

Plot (history, 'loss')



Plot (history, 'accuracy')



```
Results.loc[0] = [0, '-', '-', 10, 50000, history.history["loss"][-1], history.history["Val_loss"][-1], history.history["accuracy"][-1], history.history["Val_accuracy"][-1], round(end-start, 2)]
```

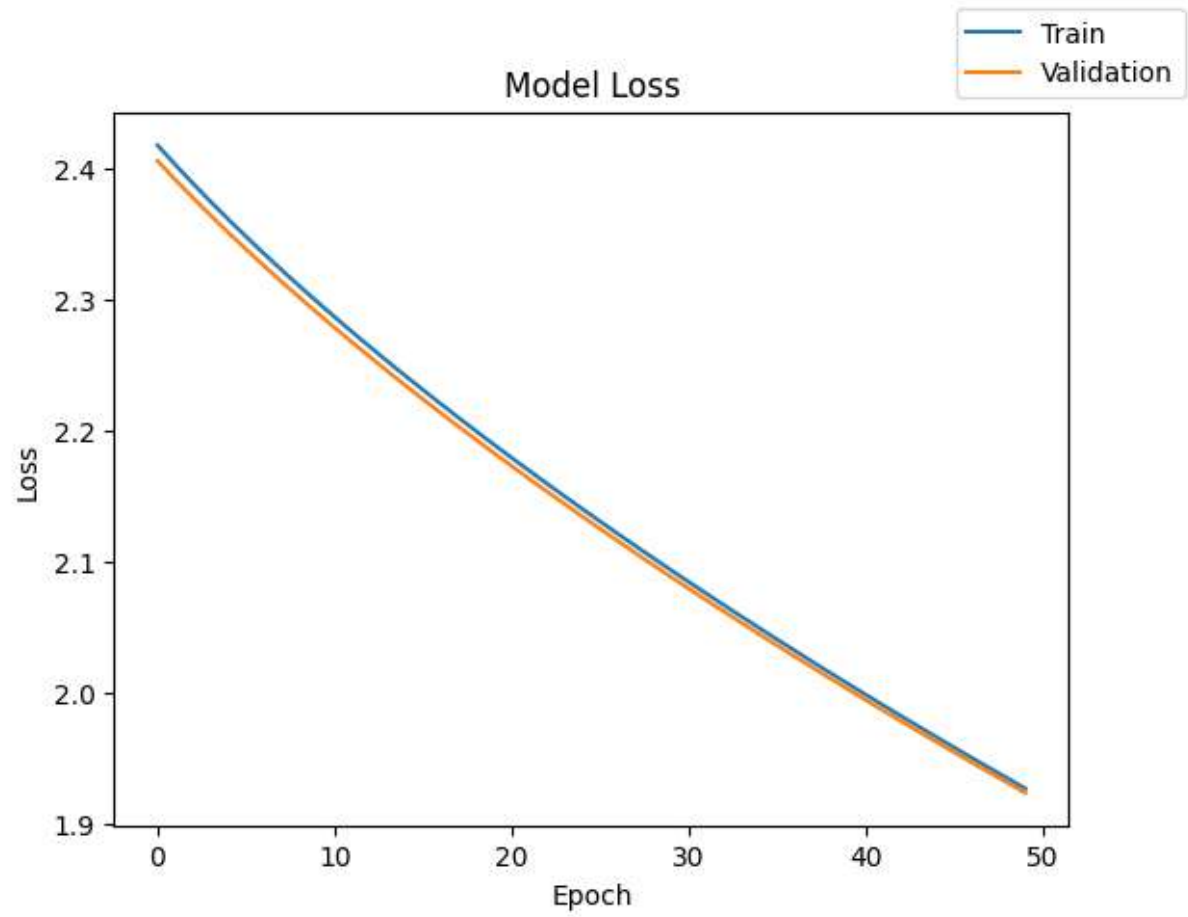
results

	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.1527	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.4306	0.4332	22.82

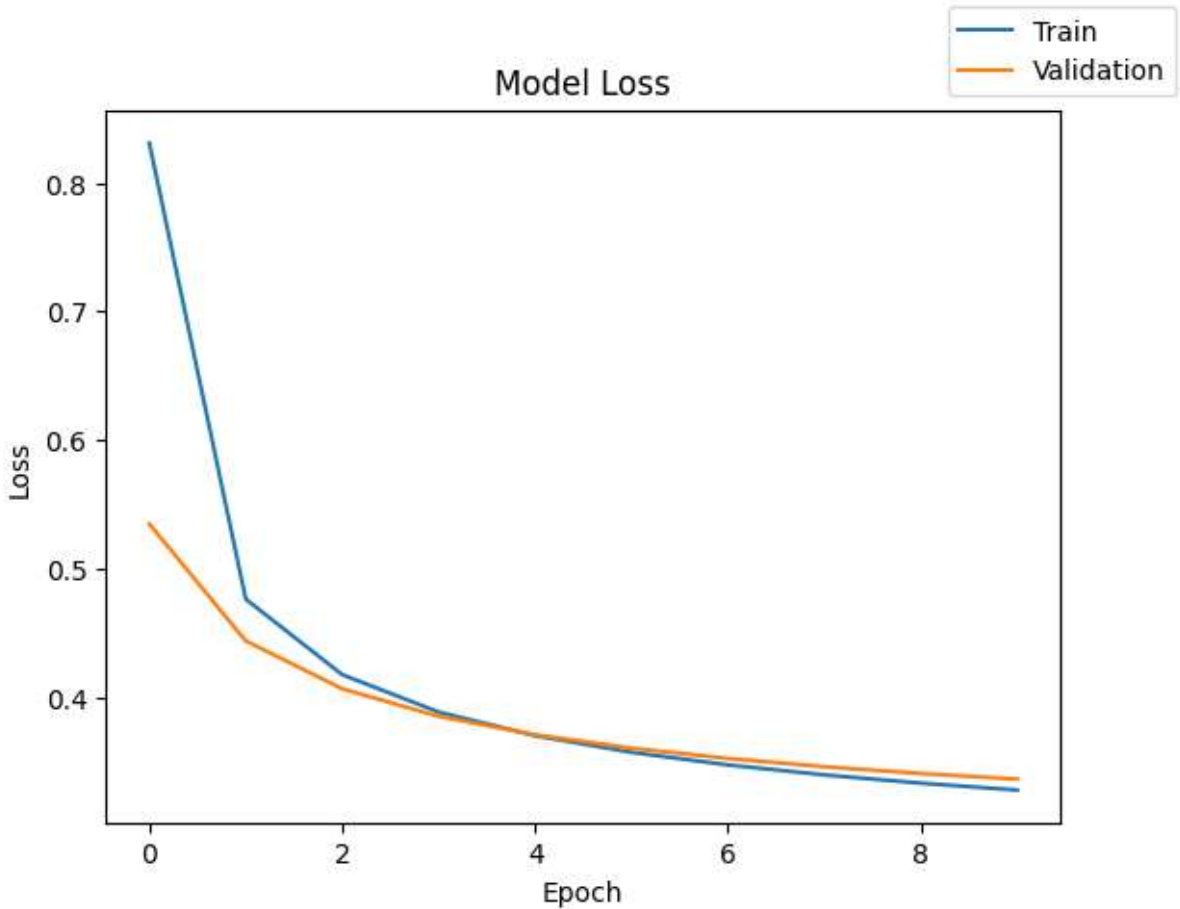
Model 2

```
Print("Time taken in seconds ", end-start)
Time taken in seconds 22.816673040390015
```

```
Plot(history, 'loss')
```



Plot (history, 'accuracy')



```
Results.loc[2] = [0, '-', '-', 10, 32, history.history["loss"][-1], history.  
history["Val_loss"][-1], history.history["accuracy"][-1], history.  
history["Val_accuracy"][-1], round(end-start, 2)]
```

Results

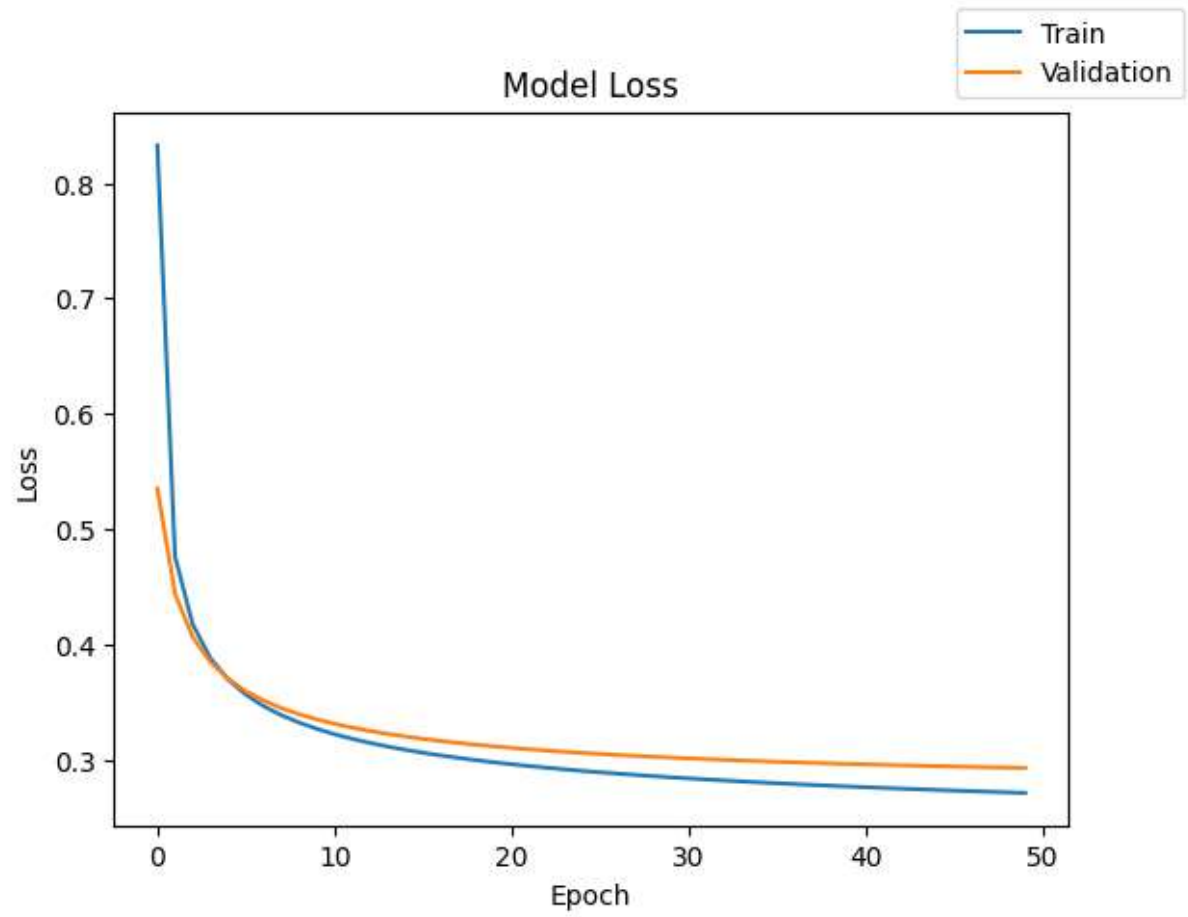
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21

Model 3

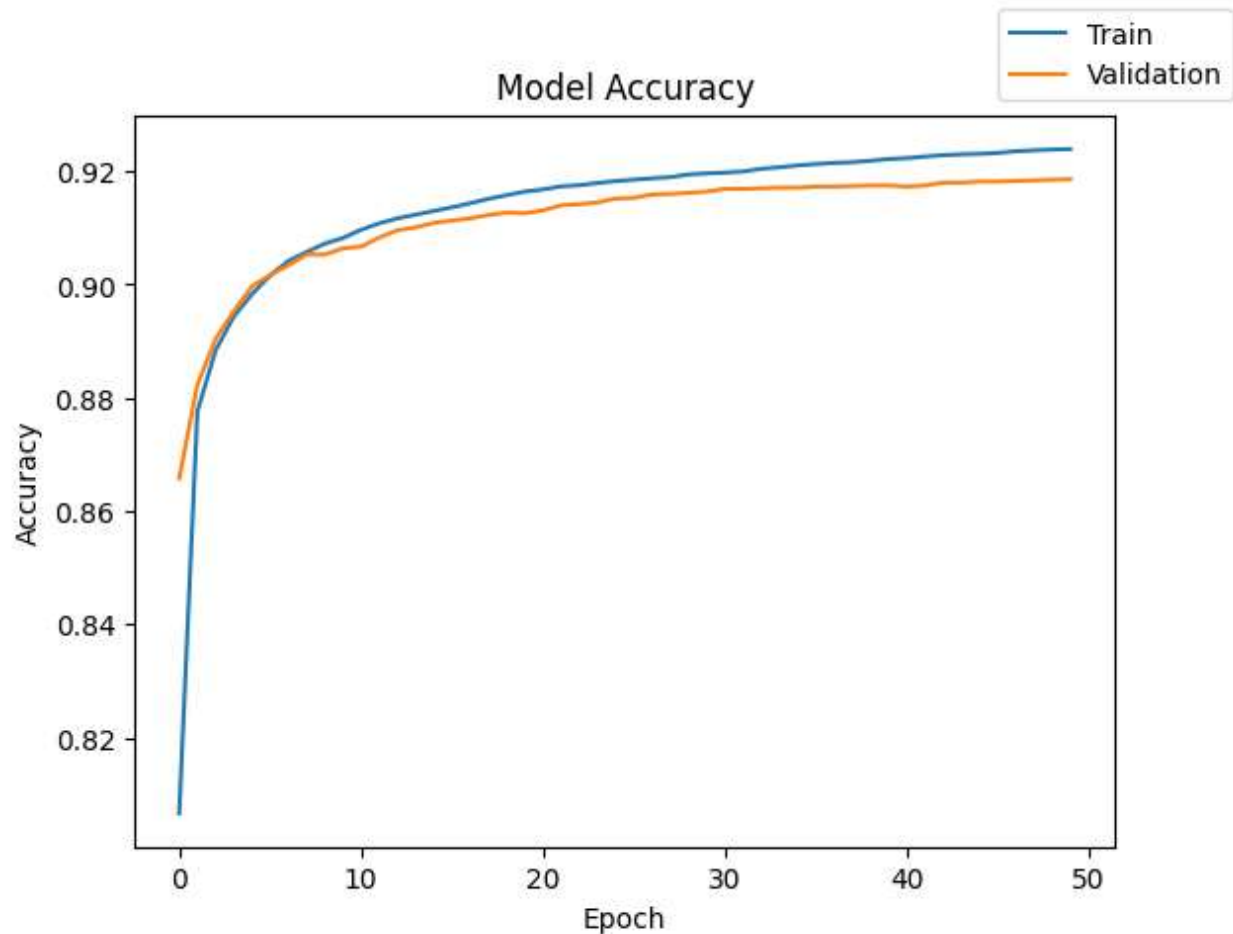
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 262.2036700248718

```
Plot (history, 'loss')
```



Plot (history, 'accuracy')



```
Results. Loc [3] = [0, '-', '-', 50, 32, history.history["loss"] [-1], history.
history ["Val_ loss"] [-1], history.history["accuracy"] [-1], history. History
["Val_ accuracy"] [-1], round (end-start,2)]
```

Results

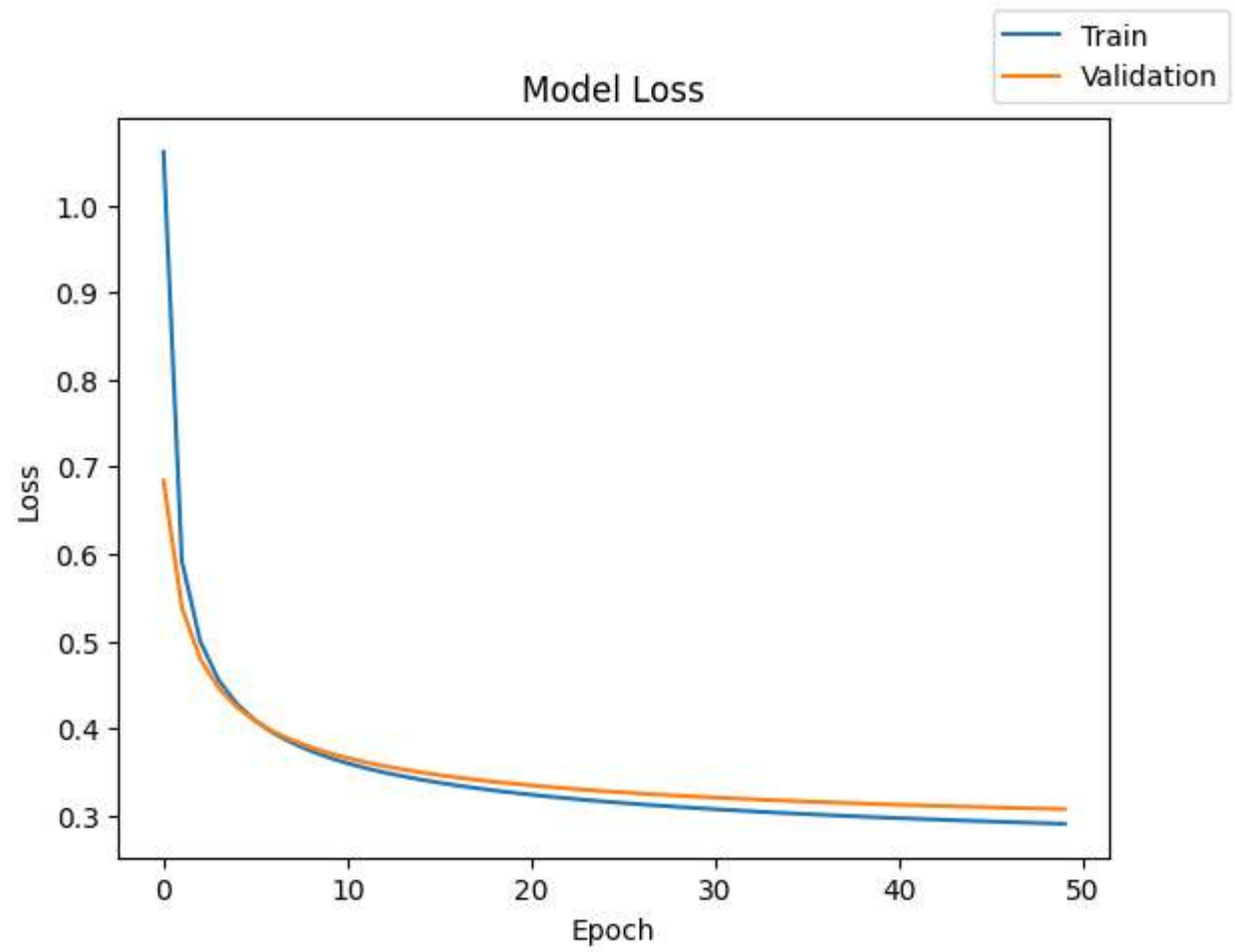
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20

Model 4

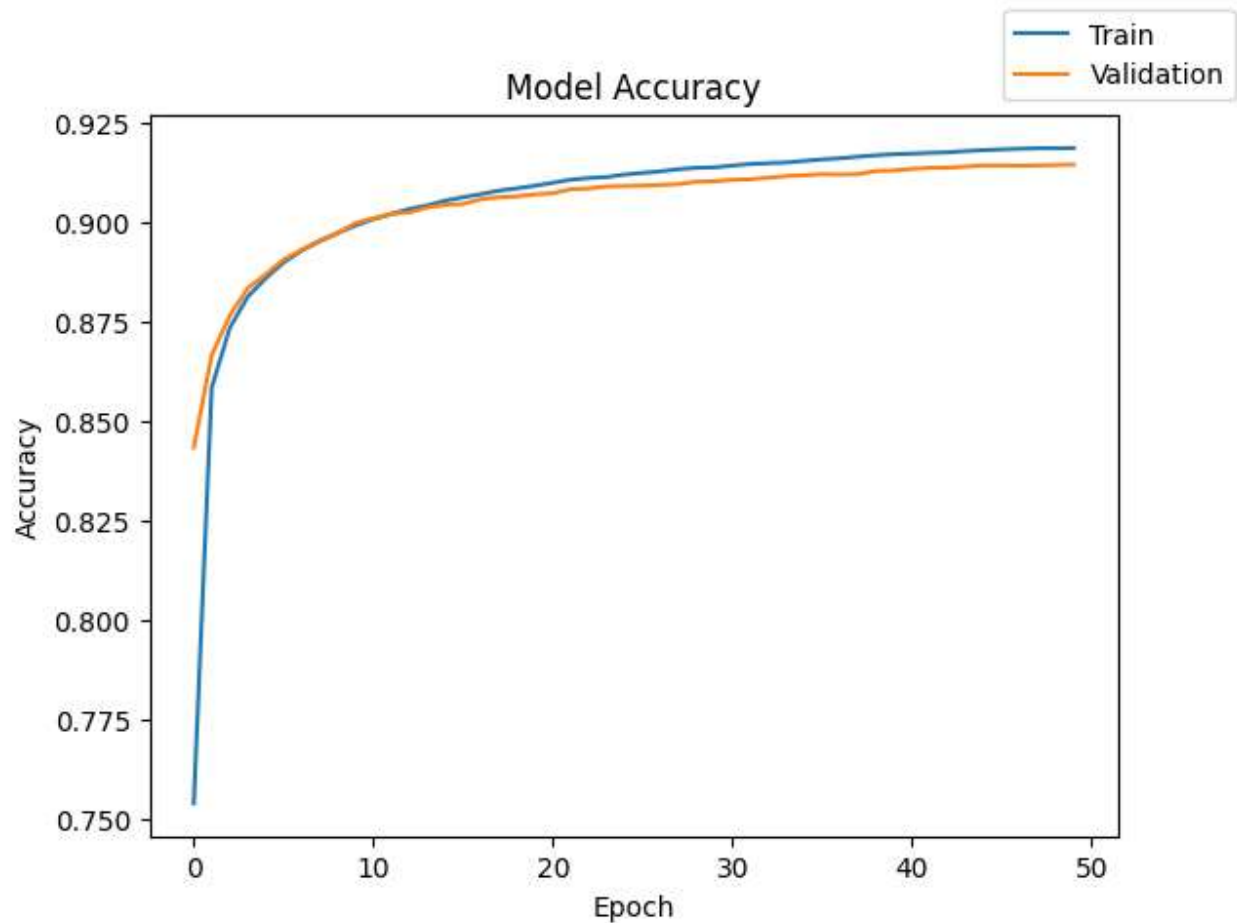
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 132.20211672782898

```
Plot (history, 'loss')
```



Plot (history, 'accuracy')



```
Results. Loc [4] = [0, '-', '-', 50, 64, history. history ["loss"] [-1], history.
history ["Val_ loss"] [-1], history. History ["accuracy"] [-1], history. History
["Val_ accuracy"] [-1], round(end-start, 2)]
```

results

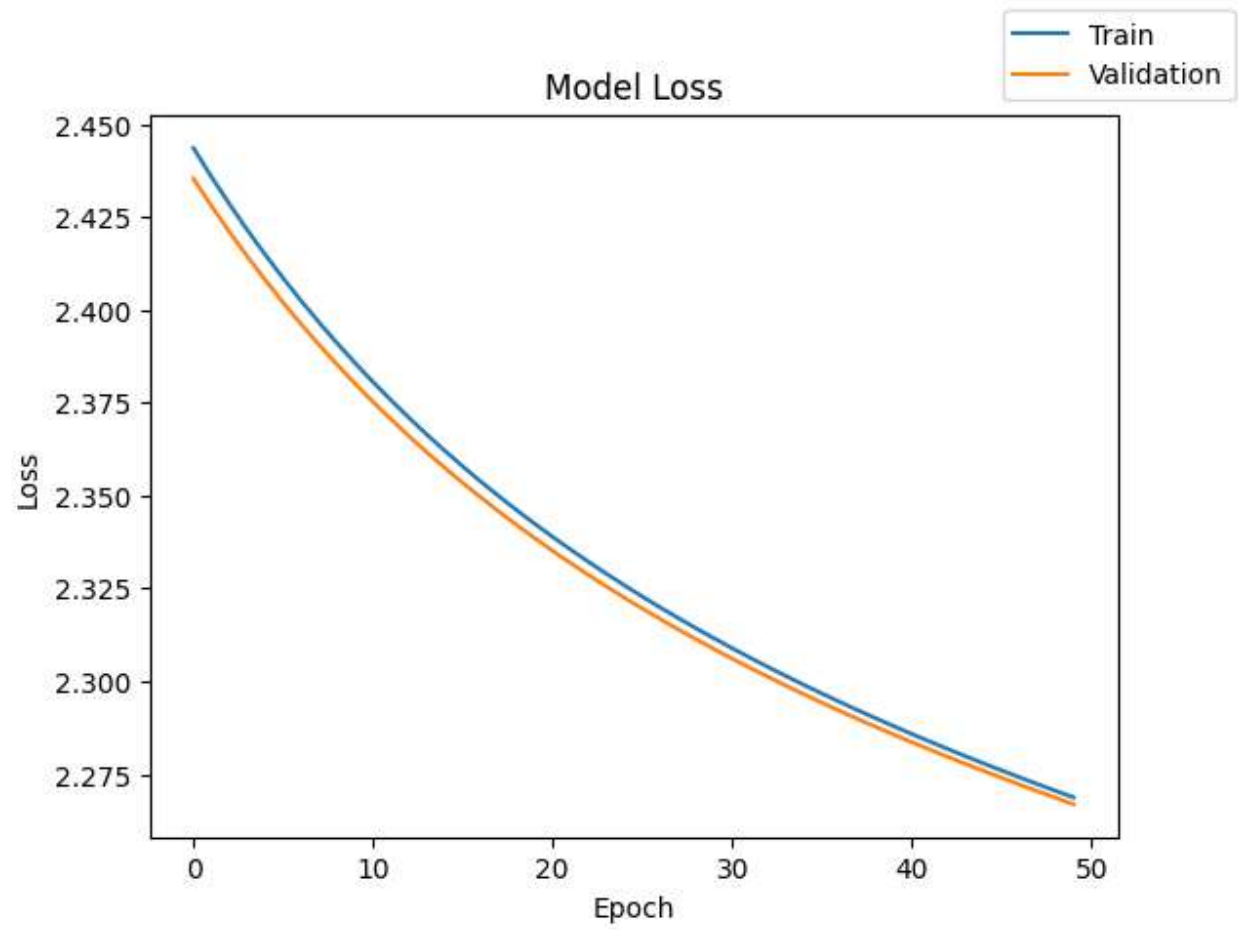
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20

Model 5

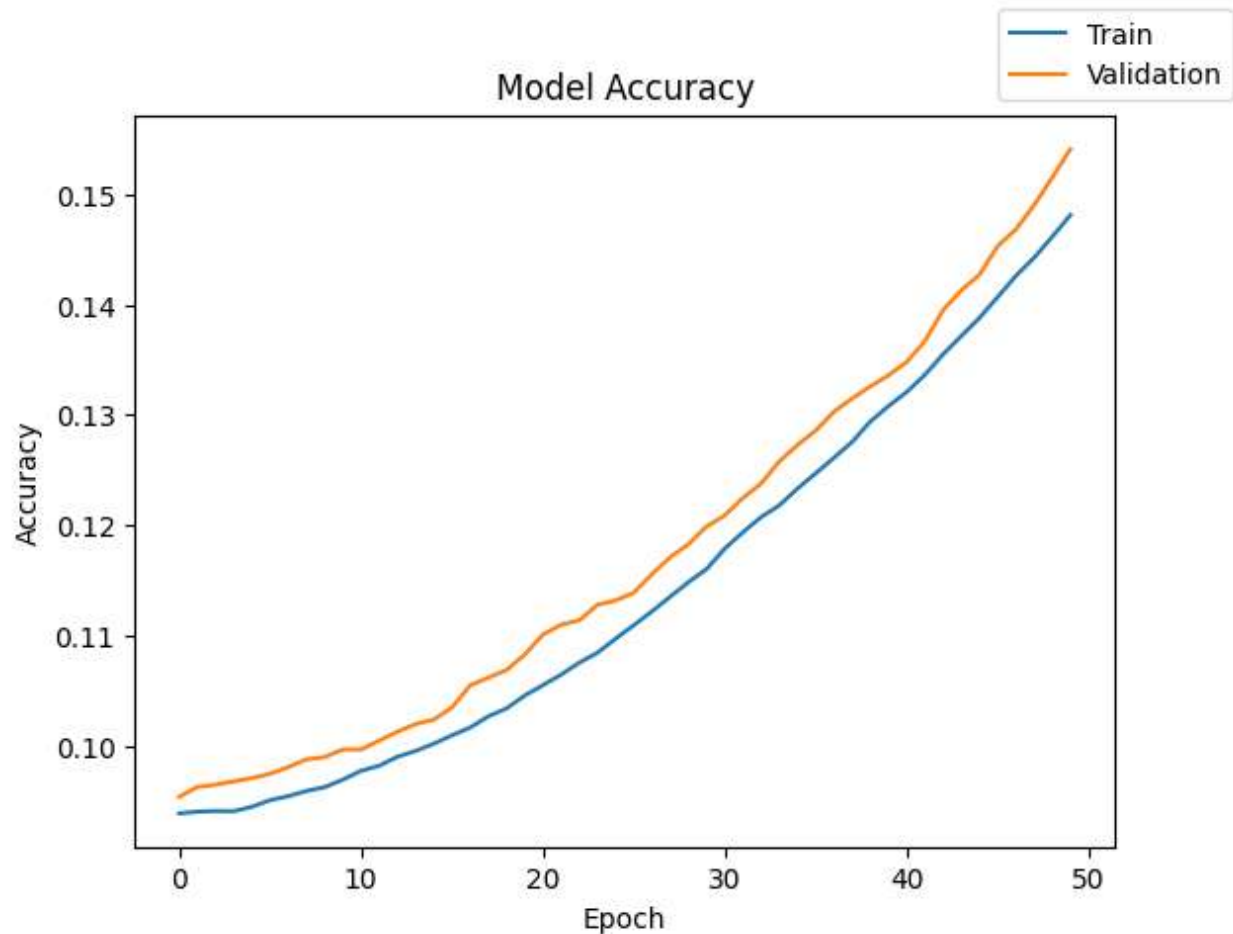
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 34.98098349

```
Plot (history, 'loss')
```



Plot (history, 'accuracy')



```
Results. Loc [5] = [1,64,'sigmoid',50,50000, history. history ["loss"] [-1],
history. History ["Val_ loss"] [-1], history. History ["accuracy"] [-1], history.
History ["Val_ accuracy"] [-1], round(end-start,2)]
```

results

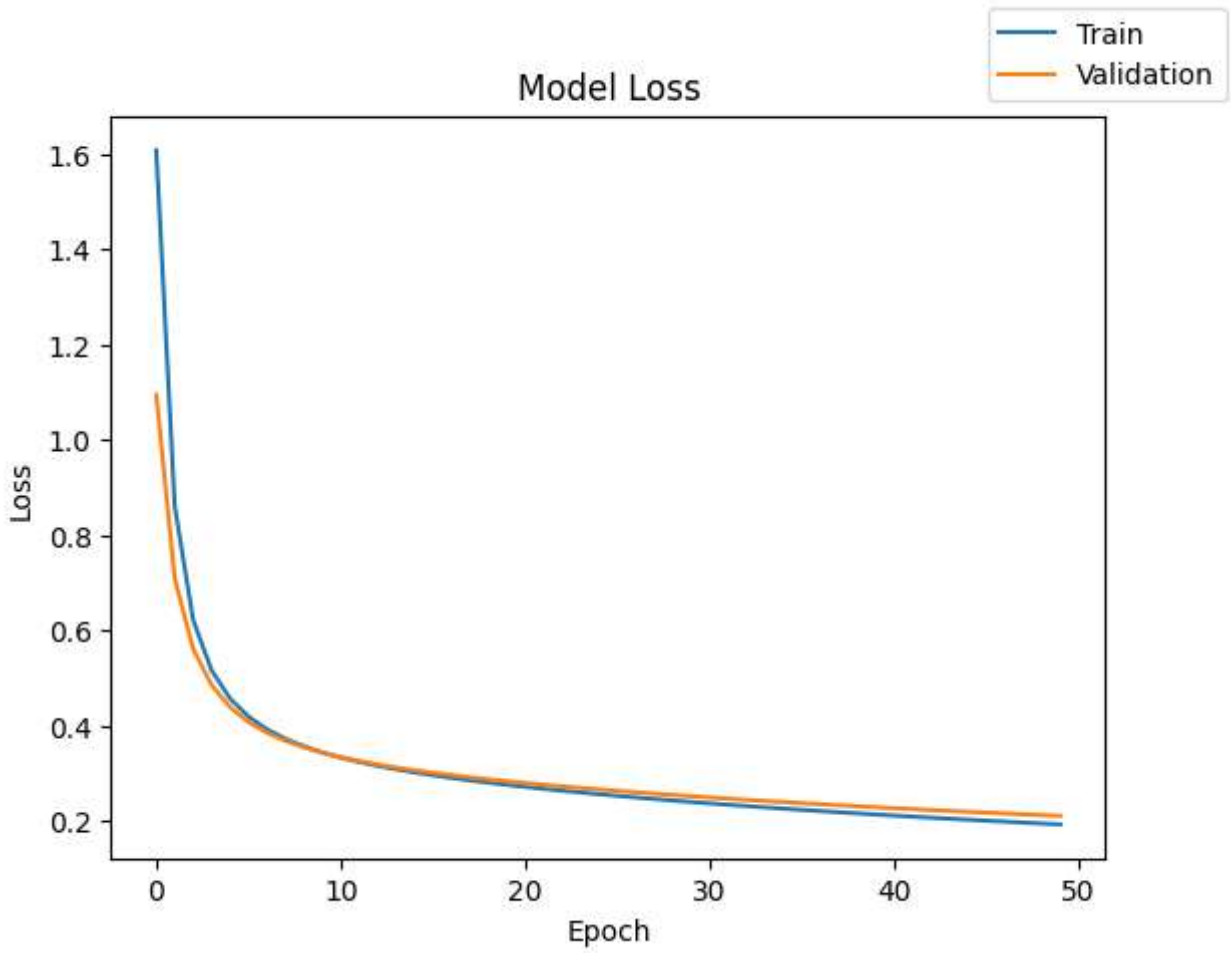
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98

Model 6

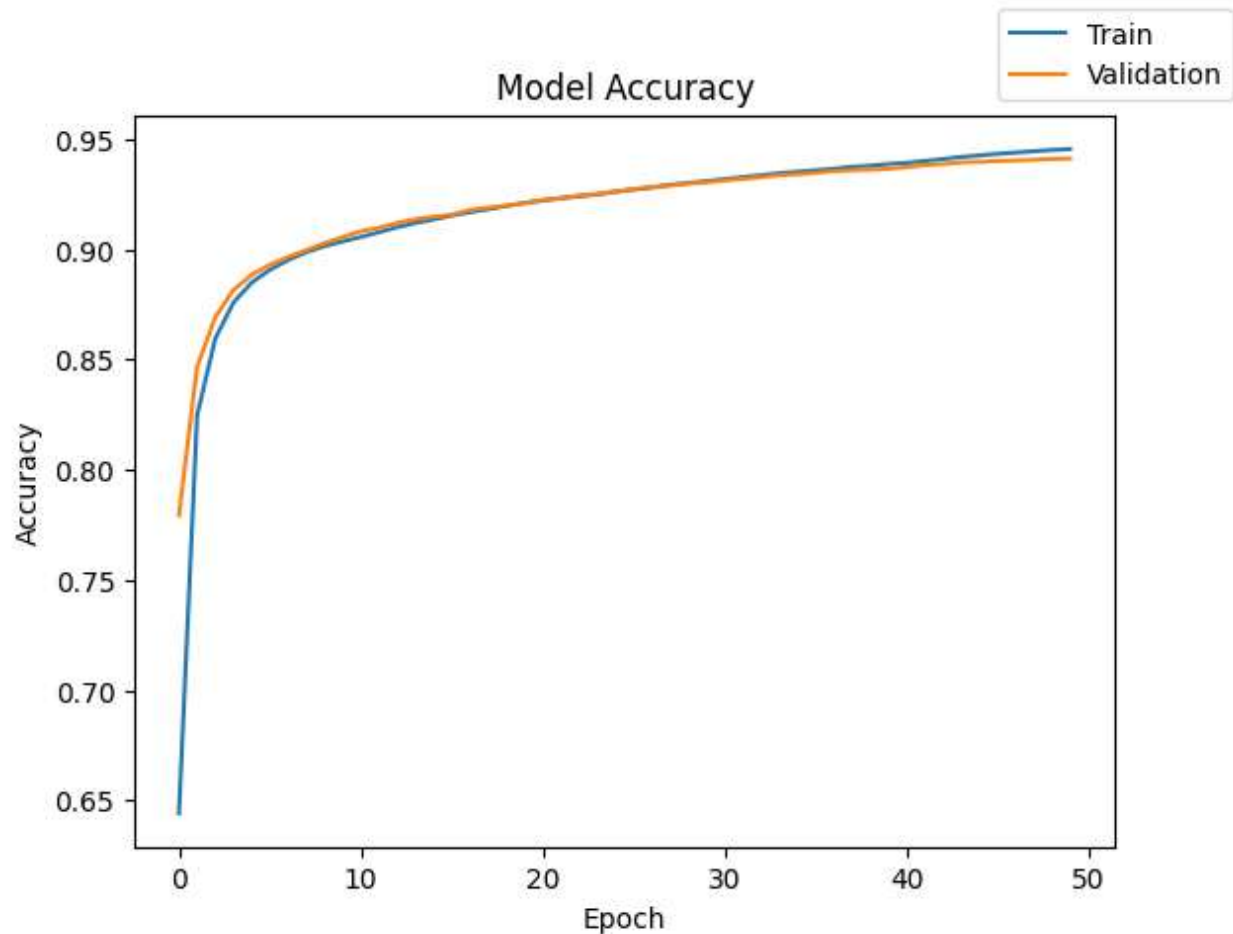
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 353.26475167274475

```
Plot (history, 'loss')
```



Plot (history, 'accuracy')



```
Results. Loc [6] = [1,64,'sigmoid',50,32, history.history["loss"][-1],
history.history["Val_loss"][-1], history.history["accuracy"][-1], history.
History["Val_accuracy"][-1], round(end-start,2)]
```

results

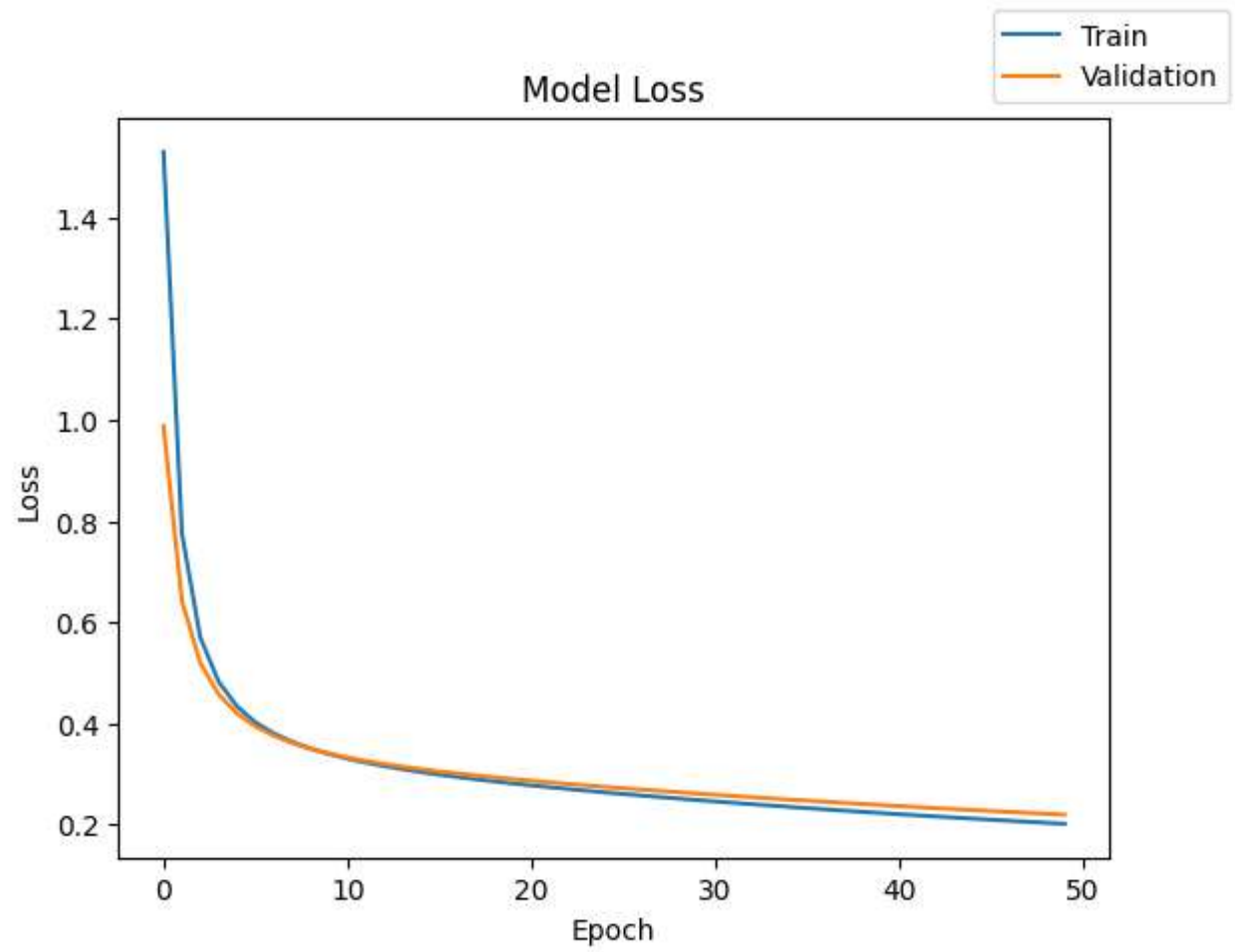
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98
6	1	64	sigmoid	50	32	0.193652	0.211530	0.94558	0.9413	353.26

Model 7

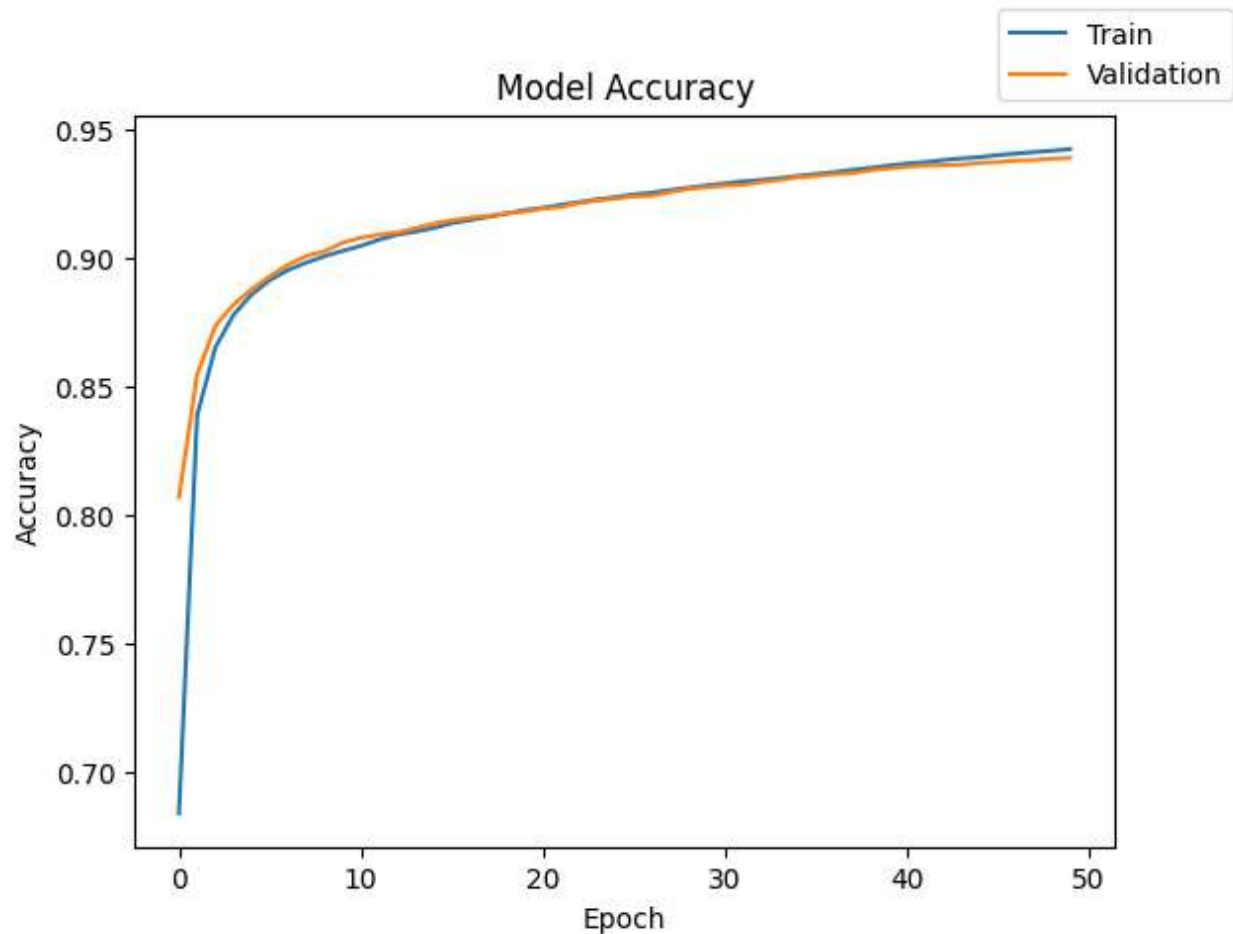
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 355.5429220199585

```
Plot (history, 'loss')
```



Plot (history, 'accuracy')



```
Results.loc [7] = [1,128,'sigmoid',50,32, history.history["loss"] [-1],
history.history ["Val_ loss"] [-1], history.history ["accuracy"] [-1], history.
history ["Val_ accuracy"] [-1], round (end-start,2)]
```

Results

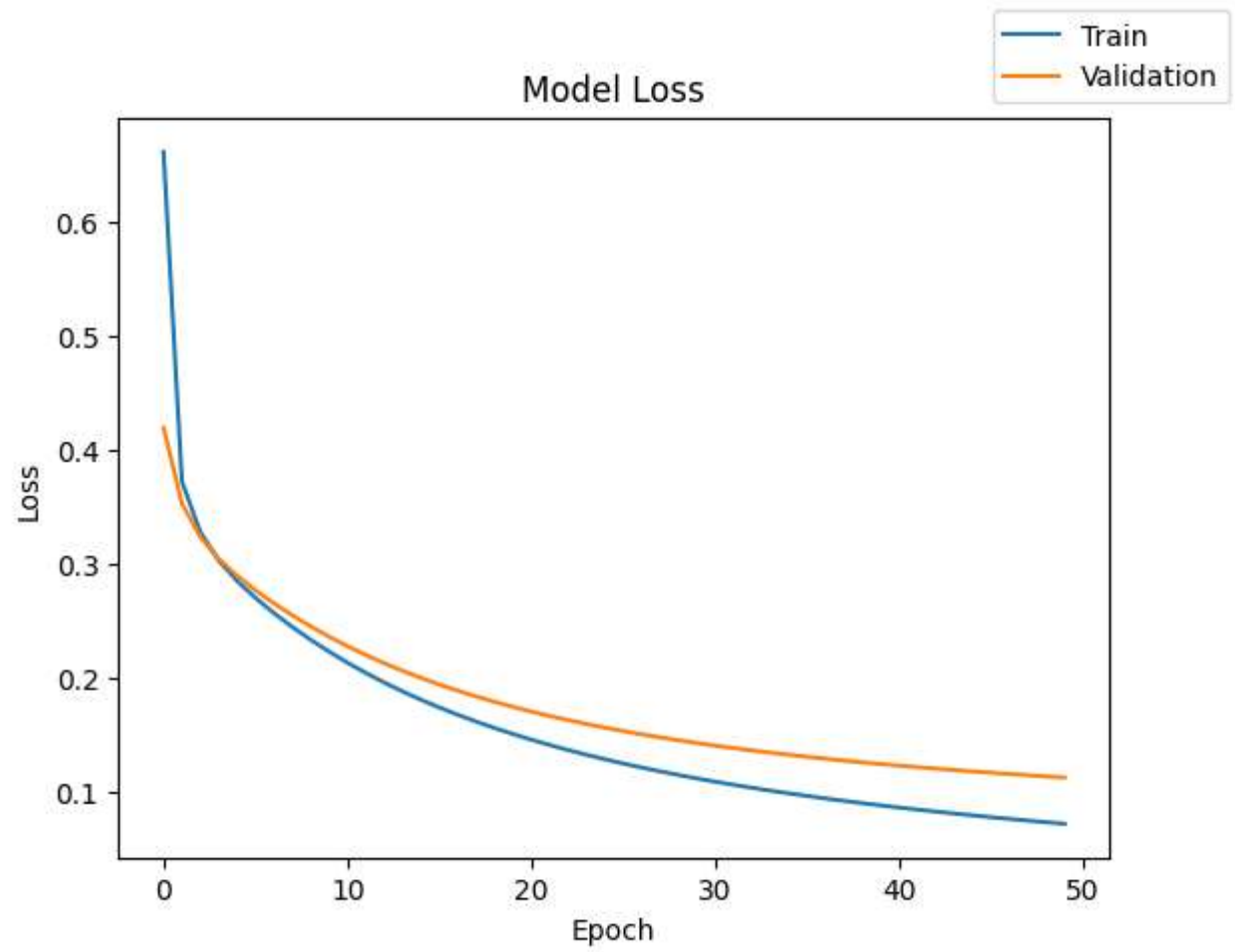
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98
6	1	64	sigmoid	50	32	0.193652	0.211530	0.94558	0.9413	353.26
7	1	128	sigmoid	50	32	0.201664	0.219673	0.94258	0.9391	355.54

Model 8

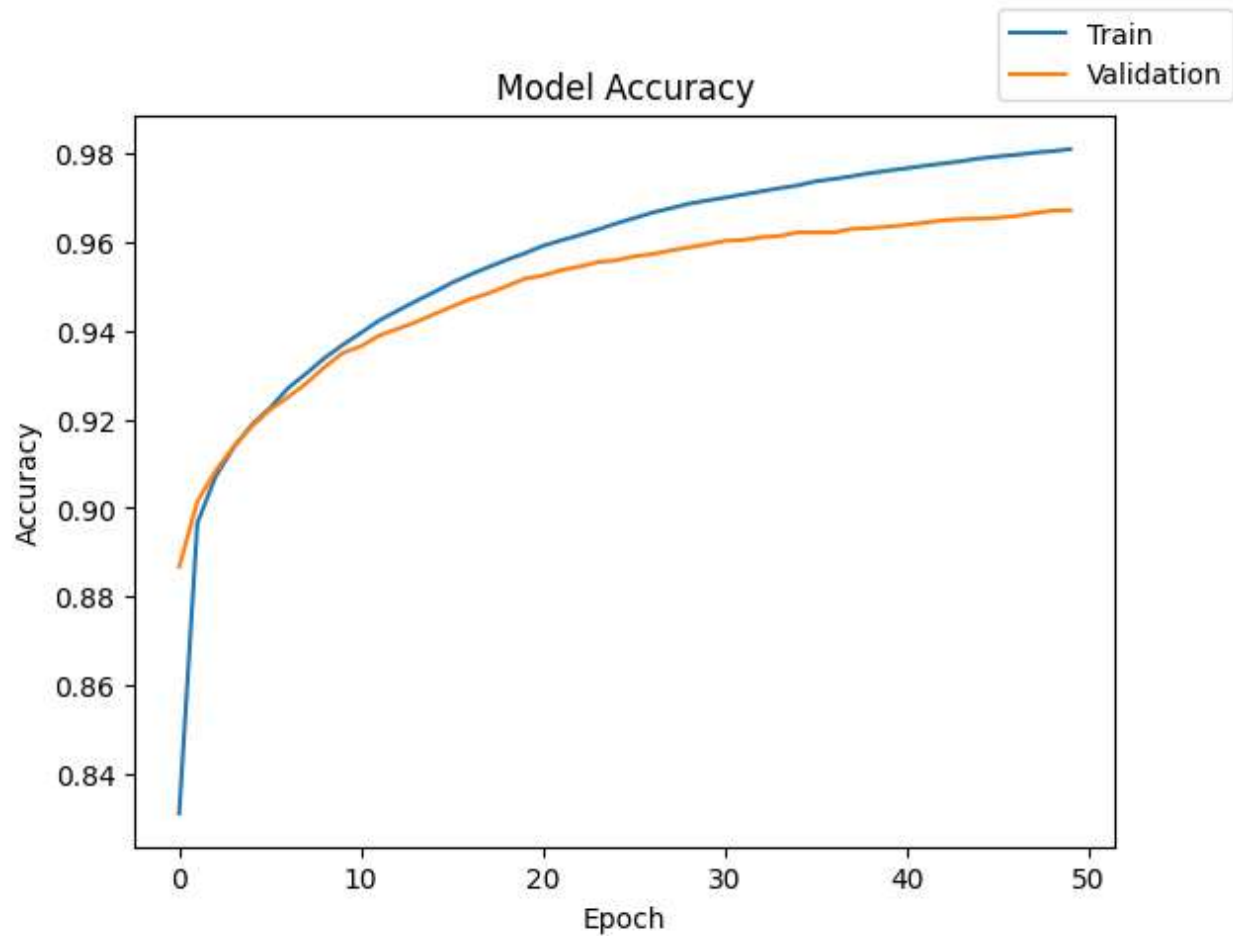
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 370.856098651886

```
Plot (history, 'loss')
```



Plot (history, 'accuracy')



```
Results. Loc [8] = [1,128,'tanh',50,32, history. History ["loss"] [-1], history.
History ["Val_ loss"] [-1], history. History ["accuracy"] [-1], history. History
["Val_ accuracy"] [-1], round (end-start,2)]
```

Results

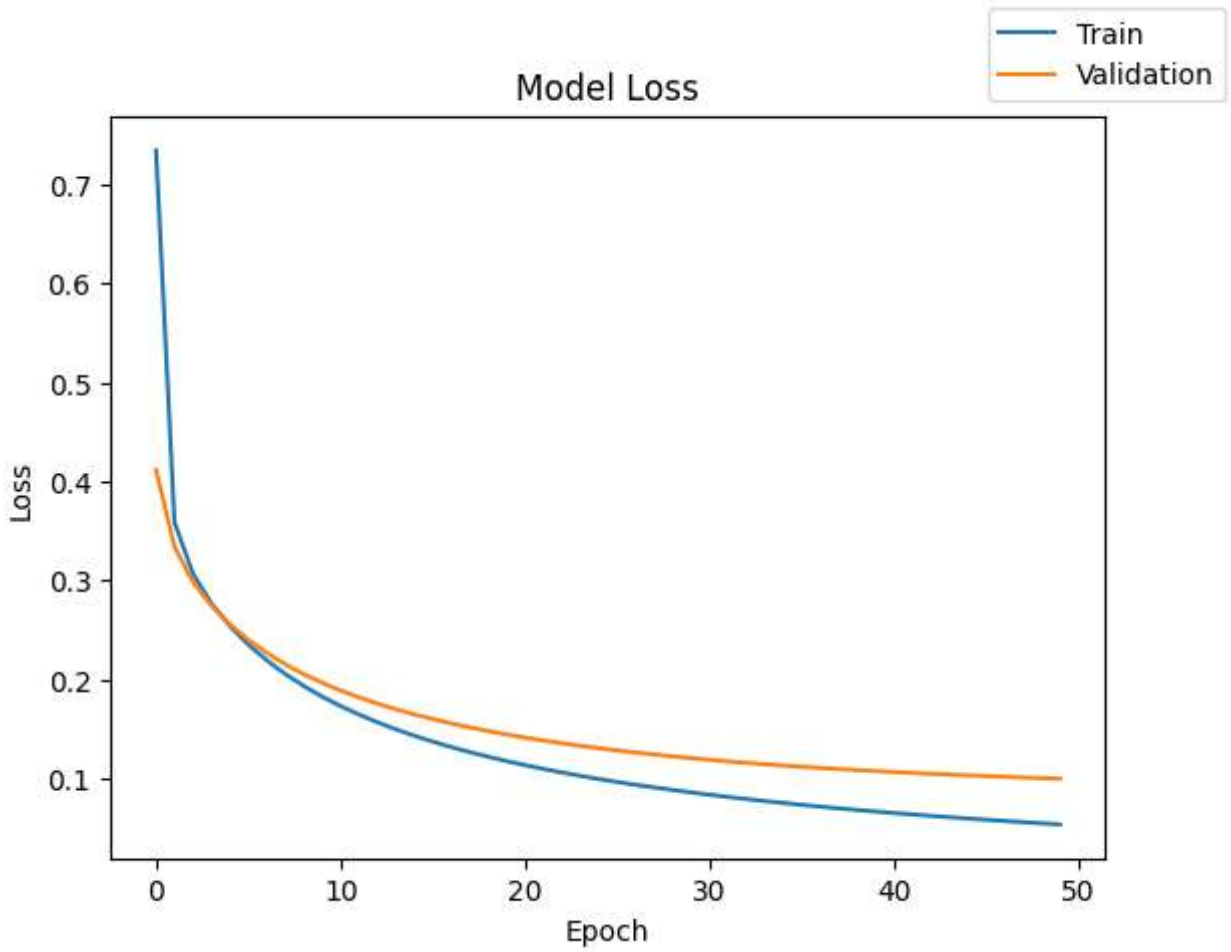
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98
6	1	64	sigmoid	50	32	0.193652	0.211530	0.94558	0.9413	353.26
7	1	128	sigmoid	50	32	0.201664	0.219673	0.94258	0.9391	355.54
8	1	128	tanh	50	32	0.072019	0.112645	0.98074	0.9670	370.86

Model 9

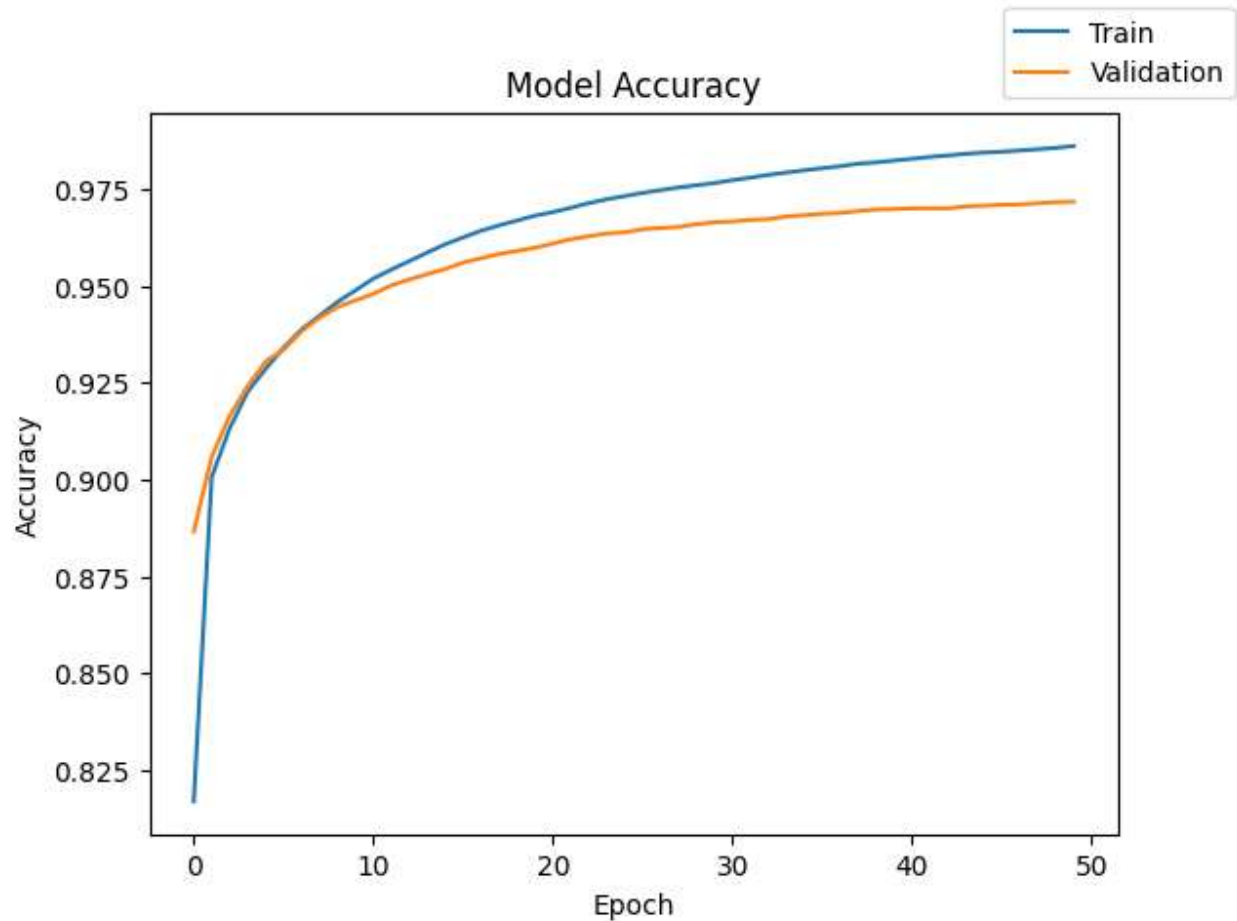
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 362.43559217453003

```
Plot (history, 'loss')
```



```
Plot (history, 'accuracy')
```



```
Results. Loc [9] = [1,128,'relu',50,32, history. History ["loss"] [-1], history.
History ["Val_ loss"] [-1], history. History ["accuracy"] [-1], history. History
["Val_ accuracy"] [-1], round (end-start,2)]
```

Results

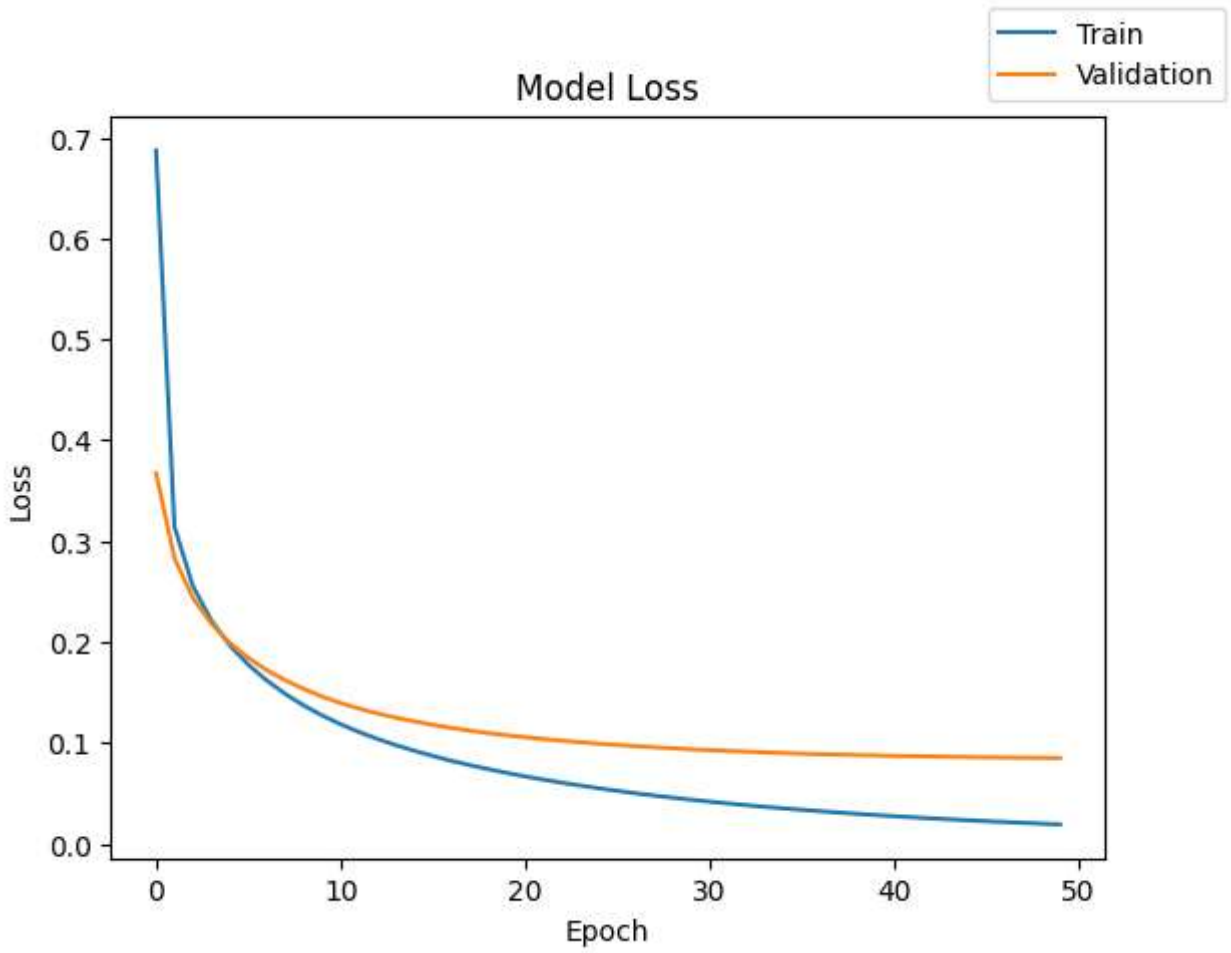
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98
6	1	64	sigmoid	50	32	0.193652	0.211530	0.94558	0.9413	353.26
7	1	128	sigmoid	50	32	0.201664	0.219673	0.94258	0.9391	355.54
8	1	128	tanh	50	32	0.072019	0.112645	0.98074	0.9670	370.86
9	1	128	relu	50	32	0.053729	0.099890	0.98620	0.9718	362.44

Model 10

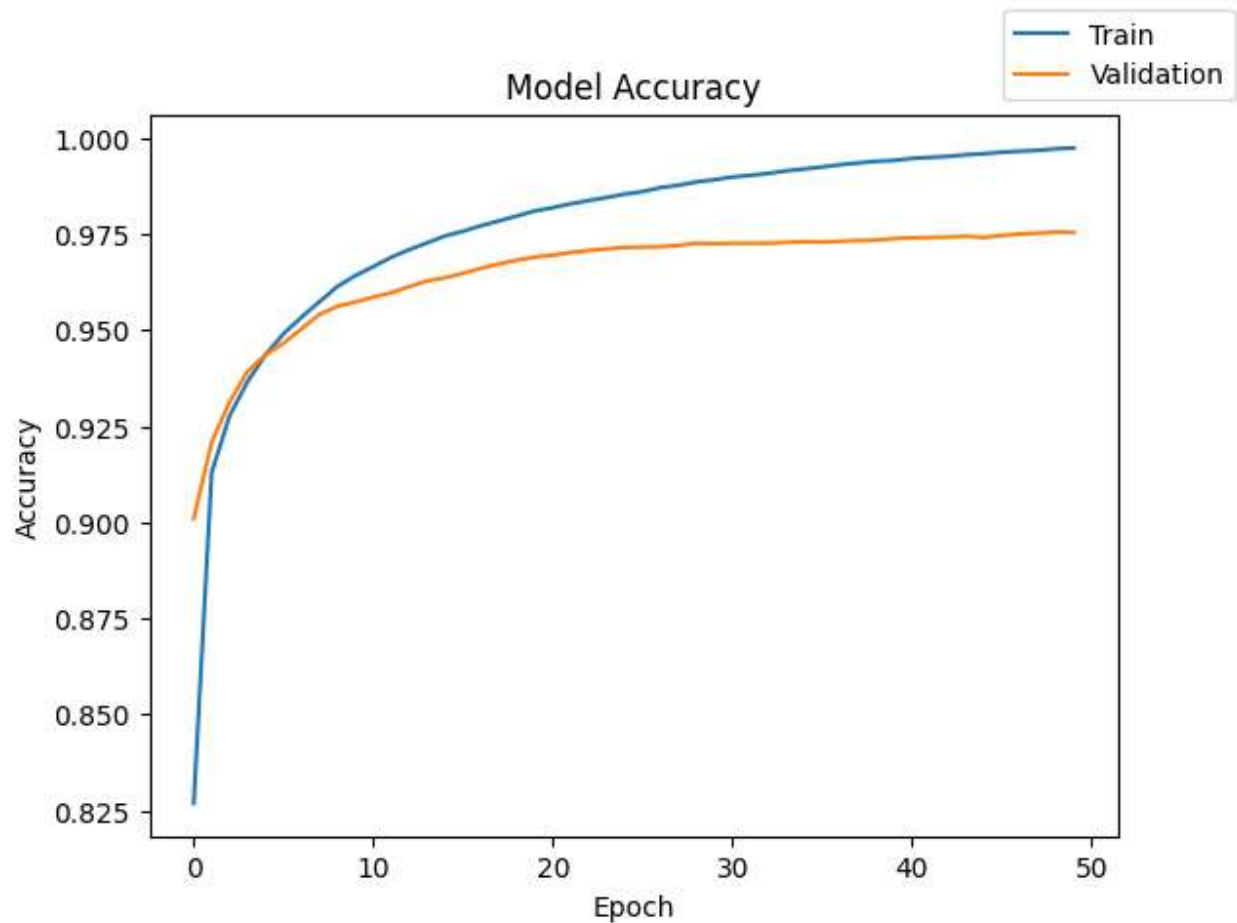
```
Print ("Time taken in seconds ", end-start)
```

```
Time taken in seconds 369.35288548469543
```

```
Plot (history, 'loss')
```



Plot (history, 'accuracy')



```
Results. Loc [10] = [2, [128,64], ['relu','tanh'],50,32, history. history
["loss"] [-1], history. history ["Val_ loss"] [-1], history. history ["accuracy"]
[-1], history. history ["Val_ accuracy"] [-1], round (end-start,2)]
```

Results

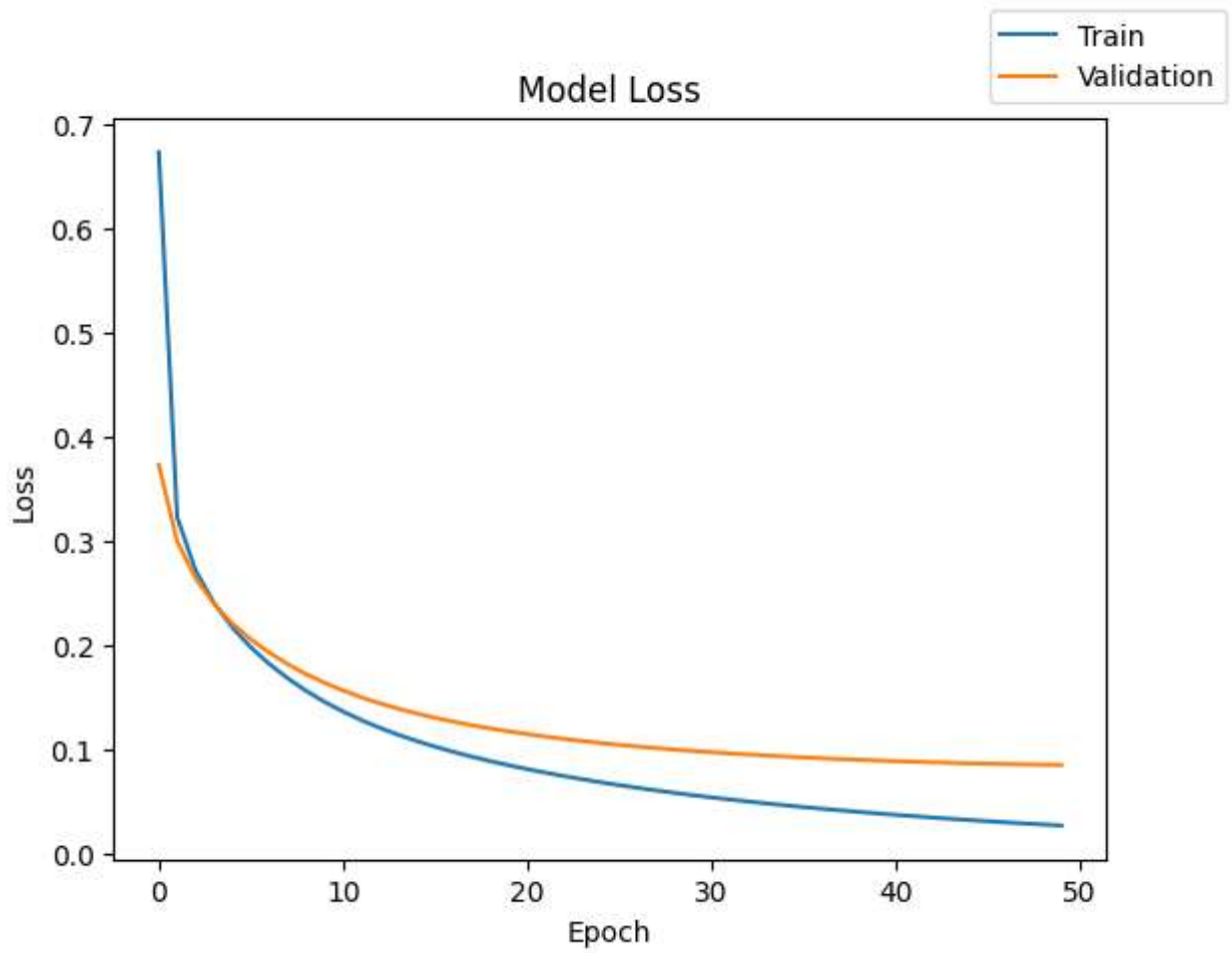
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98
6	1	64	sigmoid	50	32	0.193652	0.211530	0.94558	0.9413	353.26
7	1	128	sigmoid	50	32	0.201664	0.219673	0.94258	0.9391	355.54
8	1	128	tanh	50	32	0.072019	0.112645	0.98074	0.9670	370.86
9	1	128	relu	50	32	0.053729	0.099890	0.98620	0.9718	362.44
10	2	[128, 64]	[relu, tanh]	50	32	0.019205	0.085057	0.99736	0.9754	369.35

Model 11

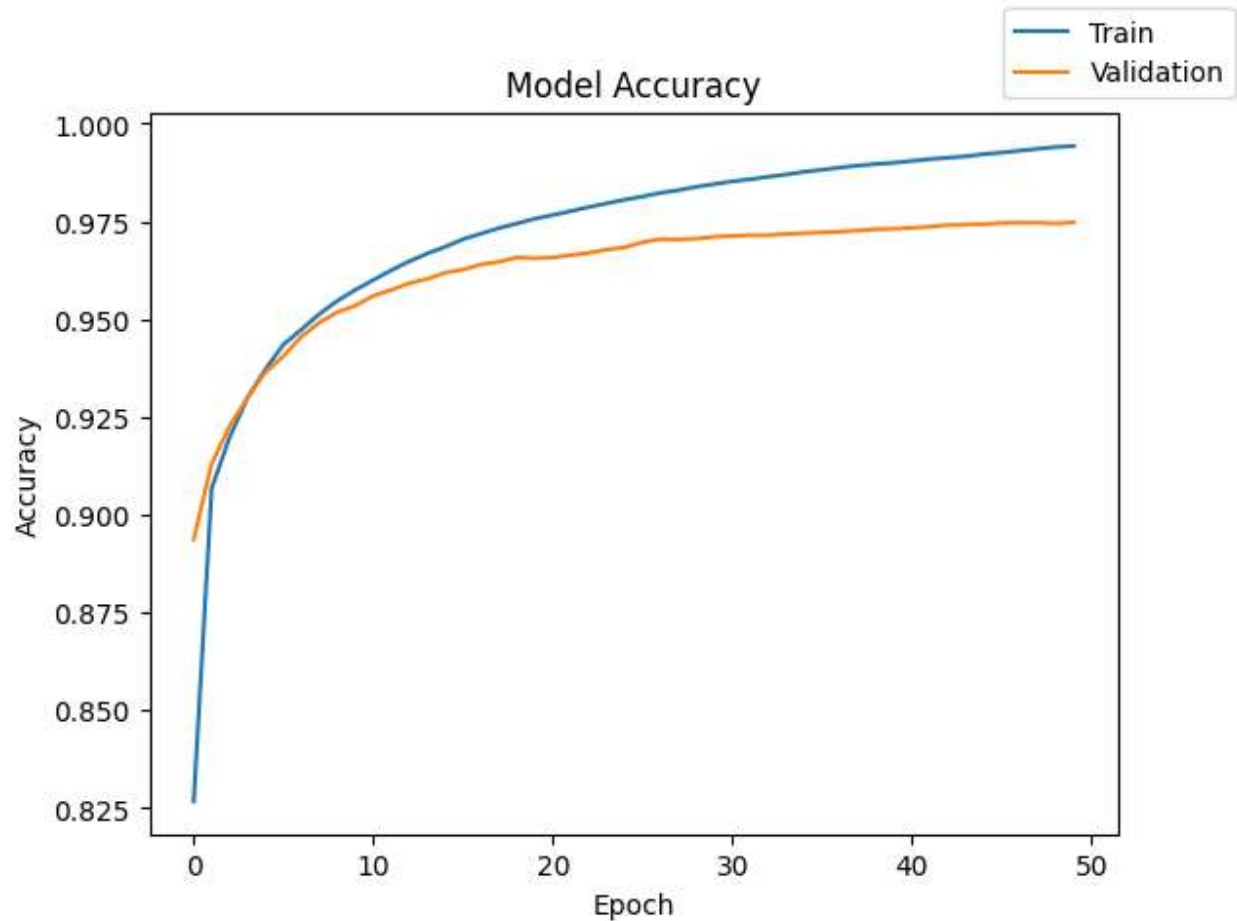
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 361.7939963340759

Plot (history, 'loss')



Plot (history, 'accuracy')



```
Results. Loc [11] = [2, [128,64], ['tanh','Relu'],50,32, history. history["loss"]
[-1], history. History ["Val_ loss"] [-1], history. history["accuracy"] [-1],
history. History ["Val_ accuracy"] [-1], round(end-start,2)]
```

results

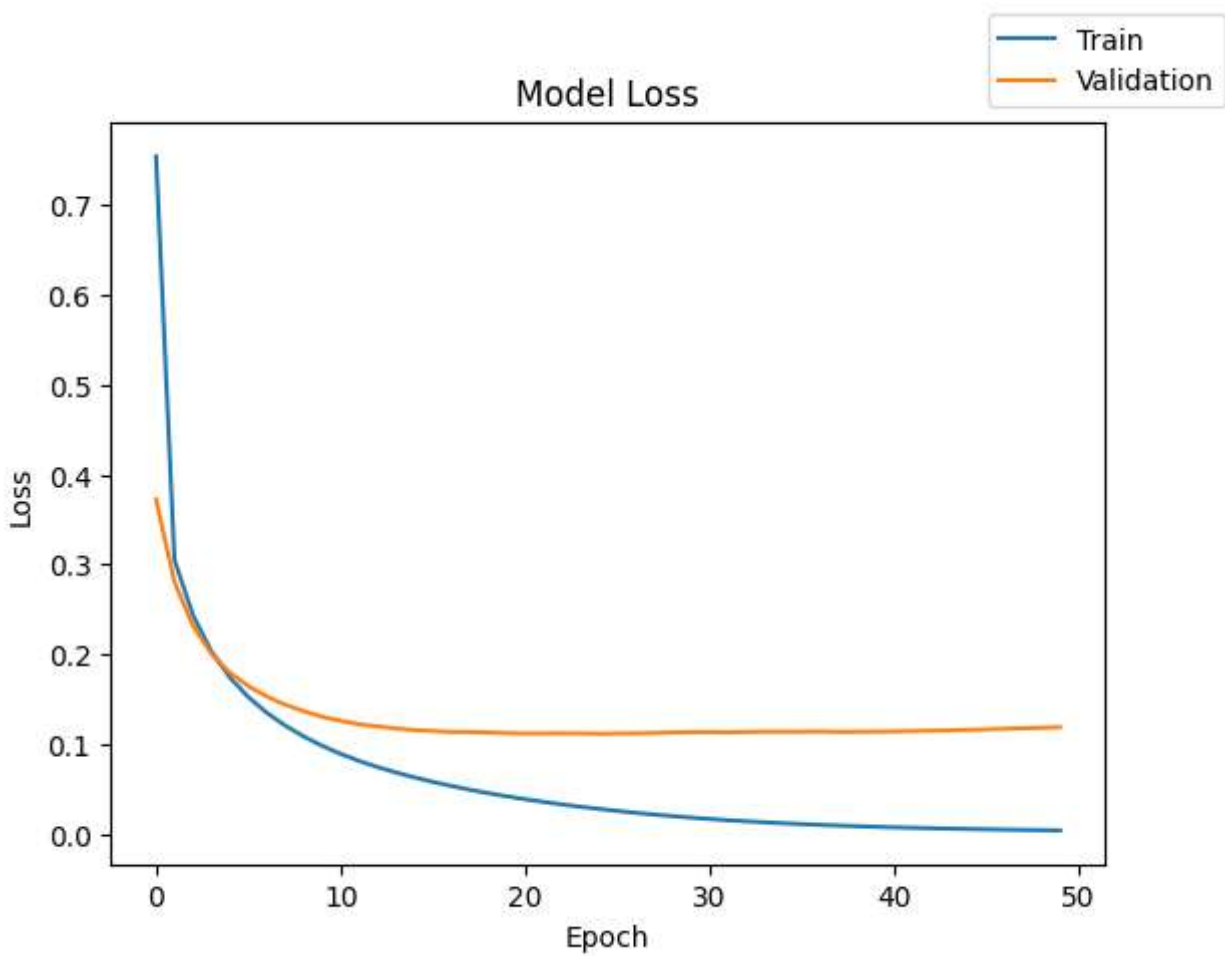
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98
6	1	64	sigmoid	50	32	0.193652	0.211530	0.94558	0.9413	353.26
7	1	128	sigmoid	50	32	0.201664	0.219673	0.94258	0.9391	355.54
8	1	128	tanh	50	32	0.072019	0.112645	0.98074	0.9670	370.86
9	1	128	relu	50	32	0.053729	0.099890	0.98620	0.9718	362.44
10	2	[128, 64]	[relu, tanh]	50	32	0.019205	0.085057	0.99736	0.9754	369.35
11	2	[128, 64]	[tanh, relu]	50	32	0.027755	0.085829	0.99430	0.9748	361.79

Model 12

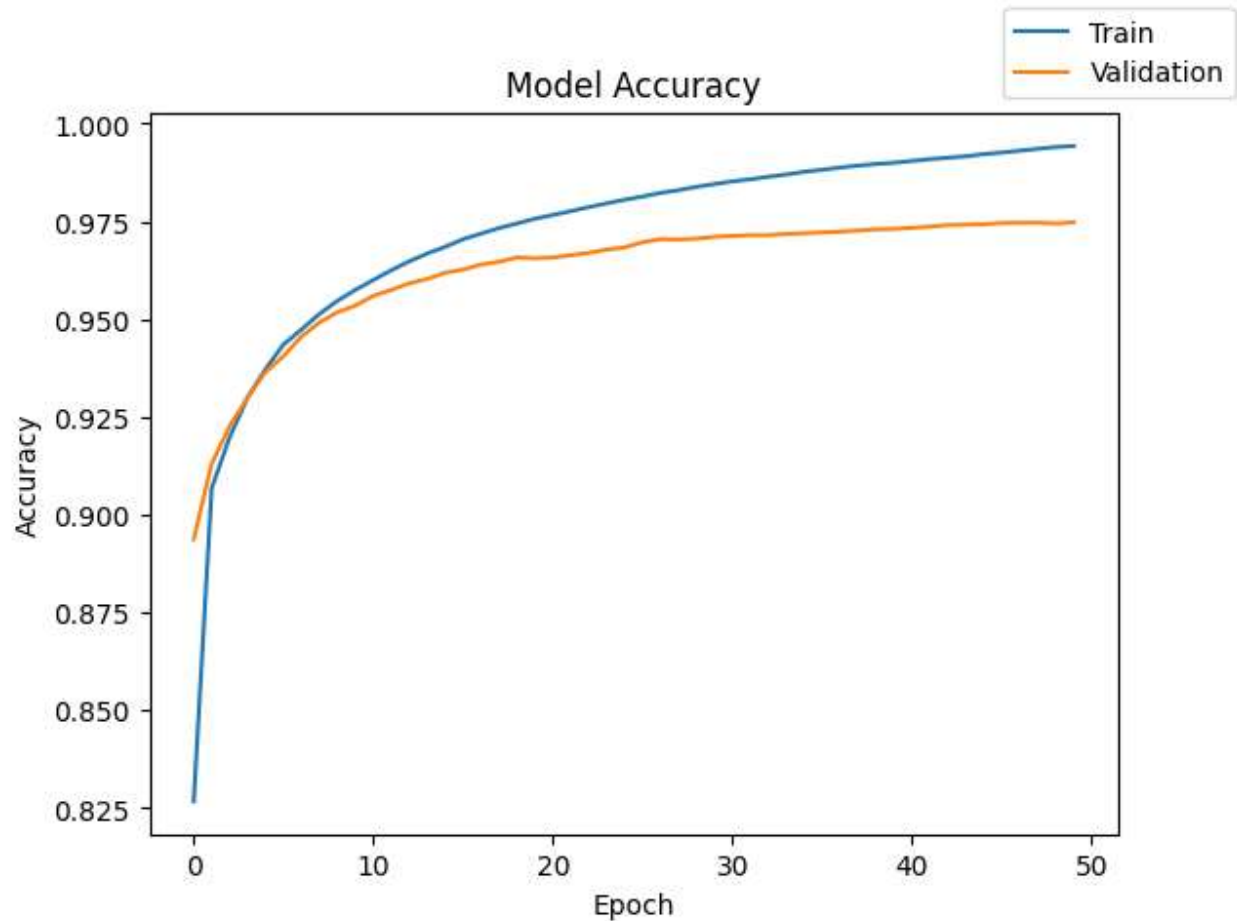
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 366.4935221672058

Plot (history, 'loss')



Plot (history, 'accuracy')



```
Results. Loc [11] = [2, [128,64], ['tanh','relu'],50,32, history. history["loss"]
[-1], history. History ["Val_ loss"] [-1], history. history["accuracy"] [-1],
history. History ["Val_ accuracy"] [-1], round(end-start,2)]
```

results

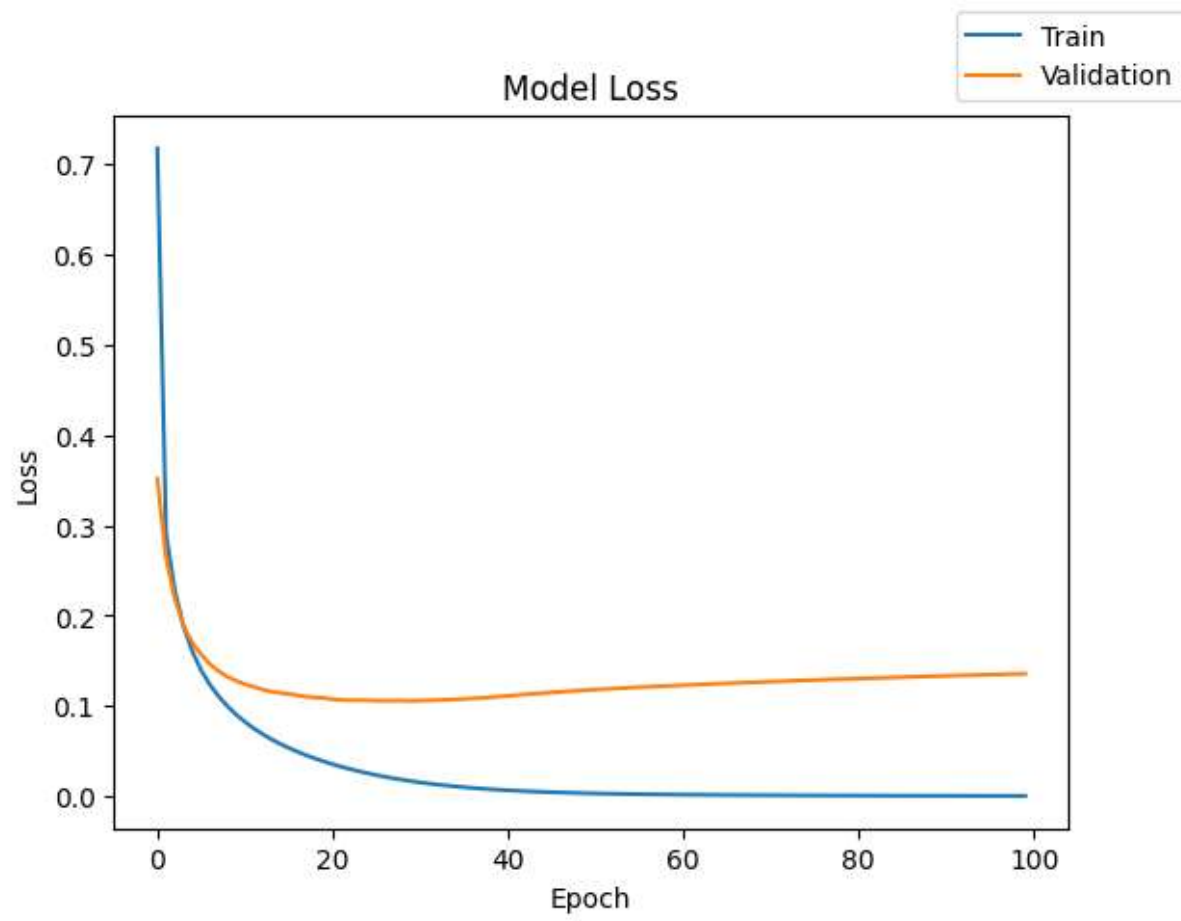
	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98
6	1	64	sigmoid	50	32	0.193652	0.211530	0.94558	0.9413	353.26
7	1	128	sigmoid	50	32	0.201664	0.219673	0.94258	0.9391	355.54
8	1	128	tanh	50	32	0.072019	0.112645	0.98074	0.9670	370.86
9	1	128	relu	50	32	0.053729	0.099890	0.98620	0.9718	362.44
10	2	[128, 64]	[relu, tanh]	50	32	0.019205	0.085057	0.99736	0.9754	369.35
11	2	[128, 64]	[tanh, relu]	50	32	0.027755	0.085829	0.99430	0.9748	361.79
12	3	[128, 64, 32]	[relu, relu, relu]	50	32	0.004095	0.118971	0.99984	0.9744	366.49

Model 13

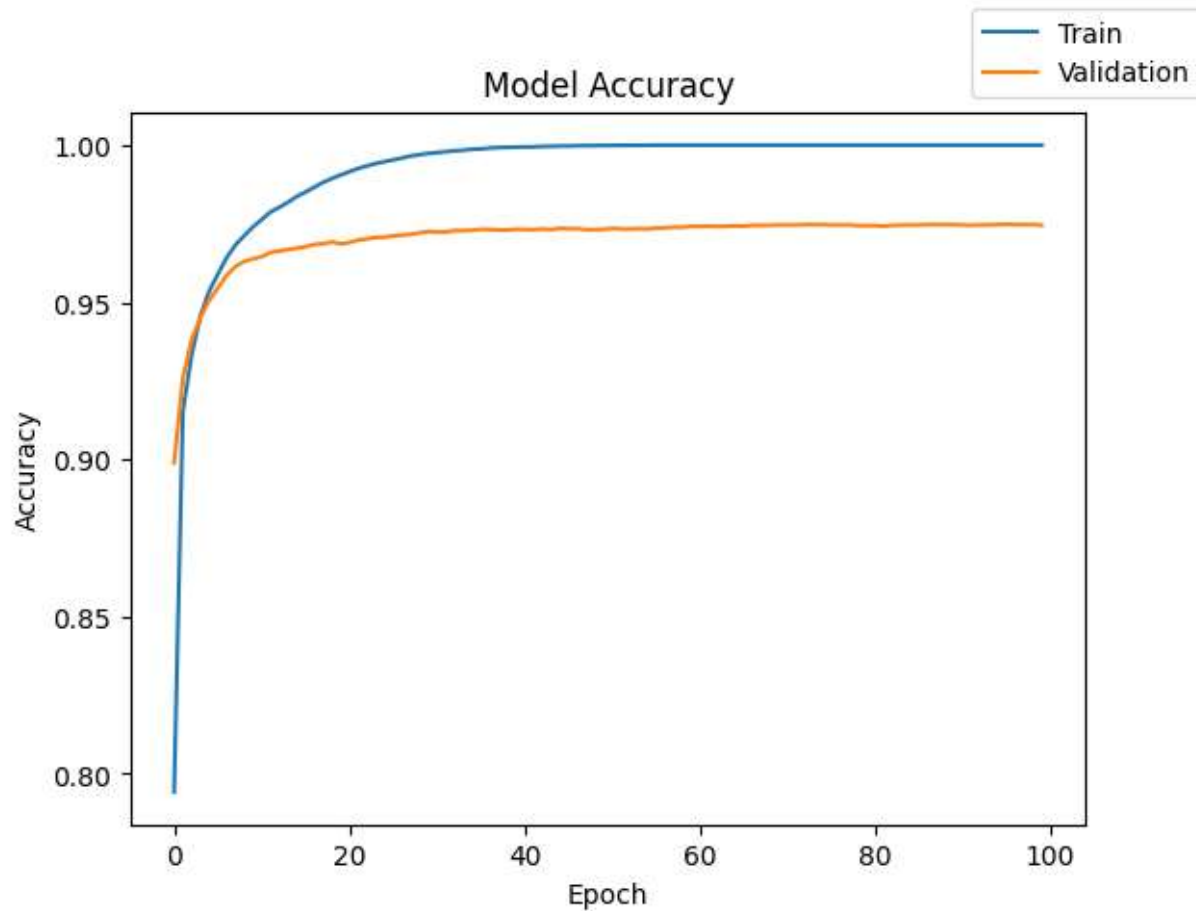
```
Print ("Time taken in seconds ", end-start)
```

Time taken in seconds 761.8118953704834

Plot (history, 'loss')



Plot (history, 'accuracy')



```
Results. Loc [13] = [3, [128,64,32], ['relu','relu','relu'],100,32, history.
history["loss"] [-1], history. History ["Val_ loss"] [-1], history.
history["accuracy"] [-1], history. History ["Val_ accuracy"] [-1], round(end-
start,2)]
```

Results

	# hidden layers	# neurons - hidden layer	activation function - hidden layer	# epochs	batch size	train loss	validation loss	train accuracy	validation accuracy	time (secs)
0	0	-	-	10	50000	2.255868	2.238811	0.15270	0.1628	7.03
1	0	-	-	50	50000	1.927448	1.924149	0.43060	0.4332	22.82
2	0	-	-	10	32	0.327848	0.336322	0.90828	0.9071	51.21
3	0	-	-	50	32	0.272220	0.293814	0.92392	0.9186	262.20
4	0	-	-	50	64	0.290571	0.307456	0.91856	0.9144	132.20
5	1	64	sigmoid	50	50000	2.268767	2.266936	0.14812	0.1541	34.98
6	1	64	sigmoid	50	32	0.193652	0.211530	0.94558	0.9413	353.26
7	1	128	sigmoid	50	32	0.201664	0.219673	0.94258	0.9391	355.54
8	1	128	tanh	50	32	0.072019	0.112645	0.98074	0.9670	370.86
9	1	128	relu	50	32	0.053729	0.099890	0.98620	0.9718	362.44
10	2	[128, 64]	[relu, tanh]	50	32	0.019205	0.085057	0.99736	0.9754	369.35
11	2	[128, 64]	[tanh, relu]	50	32	0.027755	0.085829	0.99430	0.9748	361.79
12	3	[128, 64, 32]	[relu, relu, relu]	50	32	0.004095	0.118971	0.99984	0.9744	366.49
13	3	[128, 64, 32]	[relu, relu, relu]	100	32	0.000690	0.135910	1.00000	0.9745	761.81

